Alexandre Grothendieck

Buffalo Lectures on Topos Theory

1973

Index of Results

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Introduction

The course. From April to July 1973, Alexandre Grothendieck gave a course on topos theory, in English, at the State University of New York at Buffalo, alongside two other courses, on algebraic geometry and on algebraic groups. The sessions were recorded on magnetic tape. This document relies on the recordings published by the YouTube channel "Erwin Schulhoff" in four videos; sessions are cited by date, tape and side, as the video descriptions give them, and moments by their position in the video.

What this index is. For each result, definition or example treated in a session, the index gives one line: the statement, in today's mathematical language and in our own words, with a link to the video at the exact second where it is treated. An entry corresponds to a statement, not to a minute of the lecture. Words in quotation marks are a name or a term coined in the course, kept when it matters. When what is said is sketched or incomplete, the entry gives the correct statement, with the hypotheses it requires; a claim that is false as heard is not reproduced. A passage whose audio is still doubtful is left out. Entries marked *Discussion* state no result: they point to an exchange with the audience, in our words, where something is settled, corrected or left open.

What it is not. This index is not a transcription or a summary of the course. It leaves out the explanations, the worked examples and the digressions, follows the order of the exposition only as a succession of statements, and does not quote the course. The course as delivered is a protected oral work, whose rights belong to the rightsholders of Alexandre Grothendieck; only the recordings are authoritative, and the reader is referred to them through the links. The statements here are standard mathematical statements, written by us.

What is missing. The second video (30 May to late June 1973) is indexed from a full machine transcription of the sound, which is noisier than the caption-based transcriptions used for the other videos; the dating of its meetings follows the evidence of the text and is discussed at the head of each session. Four sides whose sound is too degraded are not indexed either (4 April, tape 1, side B; 11 May, tape 2, side A; and two sides without label or date at the end of the fourth video).

Voices, hours and dates. The tapes do not name the voices in the room. McLarty [9], who was not there, names John Duskin, who made the tapes, as the one who asks about the consistency of the universe axiom (video 4, 5:16:37), and William Lawvere as having helped to save them. He counts 33 hours of topos tapes, about 16 of introduction and 17 with a smaller group; this index covers 35 hours of video, damaged sides included. He dates the last session, with the news of Deligne's proof, to 13 July 1973, where the descriptions of the videos, which this index follows, give 12 July for the whole of video 4.

Loose notions. Several notions of the course are not defined precisely in the lectures: "hypertopos", "pseudo-topos", and "all the exactness properties of sets", which McLarty reads as the single property that arbitrary products of epimorphisms are epimorphisms. The entries say so where it matters.

1 Presheaves and sheaves on a space

Session of 4 April 1973

Video 1/4 (`URU5jgTvjhU`): tape 1, side A (0:00:00–0:49:11); tape 1, side B (0:49:11–1:36:58);

tape 2, side A (1:36:58–2:28:43); tape 2, side B (2:28:43–2:33:29).

Tape 1, side A

Proposition. *The open sets $\mathcal{O}(X)$ of a space X, ordered by inclusion, form a category:* $\mathrm{Hom}(U, V)$ *is a singleton if $U \subset V$, empty otherwise.* 0:03:26

Definition. A presheaf of sets on X is a functor $F\colon \mathcal{O}(X)^{\mathrm{op}} \to$ **Set**; the maps $F(V) \to F(U)$, $U \subset V$, are the restrictions. 0:05:35

Definition. A morphism of presheaves $F \to G$ is a natural transformation: maps $F(U) \to G(U)$ compatible with the restrictions. 0:07:46

Definition. F is separated if, for every open covering $U = \bigcup U_i$, the map $F(U) \to \prod_i F(U_i)$ is injective. 0:09:56

Definition. F is a sheaf if, for every covering $U = \bigcup U_i$, the sequence $F(U) \to \prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j)$ is exact (equalizer). 0:14:19

Proposition. *The sheaves on X form a full subcategory* $\mathrm{Sh}(X)$ *(denoted here "Top(X)") of the category of presheaves.* 0:16:28

2 Examples of sheaves

Example. For a set E, the presheaf $U \mapsto E^U$ of maps from U to E is a sheaf. 0:18:47

Definition. A subpresheaf $G \subset F$ is a family of subsets $G(U) \subset F(U)$ stable under the restrictions. 0:20:54

Proposition. *A subpresheaf of a separated presheaf is separated.* 0:24:12

Proposition. *A subpresheaf G of a sheaf F is a sheaf if and only if membership of a section in G is a local property on U.* 0:24:12

Example. If E is a topological space, the continuous maps from open sets of X to E form a subsheaf of $U \mapsto E^U$. 0:26:19

Example. For differentiable (or analytic) manifolds X and E, the differentiable (resp. analytic) maps form a sheaf. 0:27:23

Example. For $p\colon E \to X$ continuous, the continuous sections $\Gamma(U, E)$ over the open sets U form a sheaf. 0:29:29

Proposition. *The sheaf of continuous maps from X to E is isomorphic to the sheaf of sections of the second projection $E \times X \to X$.* 0:32:45

Example. On a locally compact space X, the Radon measures on the open sets of X form a sheaf. 0:34:59

Example. On a differentiable manifold, the distributions on the open sets form a sheaf. 0:36:04

Example. $U \mapsto \{\text{subsets of } U\}$, with restriction by intersection, is a sheaf; the closed subsets of U form a subsheaf. 0:36:04

Example. On an analytic space, the analytic subsets of the open sets form a sheaf. 0:38:11

3 Sheaves of sections and the case of a point

For sheaves in general he recommends Godement's book as the standard reference. [1] 0:39:18

A second book on sheaves, mainly cohomological, is Bredon's. [2] 0:40:24

Proposition. *The sheaf of sections $\widetilde{E}$ of a space E over X defines a functor* $\mathrm{Fib}(X) \to \mathrm{Sh}(X)$, $f \mapsto (s \mapsto f \circ s)$. 0:41:27

Counterexample. This functor is not fully faithful: for X a point, it is identified with the forgetful functor from topological spaces to **Set**. 0:44:40

Proposition. *On a point, the functor* $\Gamma \colon F \mapsto F(\mathrm{pt})$ *is an equivalence of categories* $\mathrm{Sh}(\mathrm{pt}) \simeq$ **Set**. 0:45:45

Definition. For U open, $\Gamma_U \colon F \mapsto F(U)$ is a functor $\mathrm{Sh}(X) \to$ **Set**; we write $\Gamma(F) = \Gamma(X, F) = F(X)$. 0:46:51

Tape 1, side B

Not indexed (sound too degraded to be transcribed).

4 Limits and colimits

Tape 2, side A

Discussion. On the word « étalé »: it comes from domaines étalés in complex analysis, has no English equivalent, and 'local homeomorphism' is judged misleading and unfit for differentiable or analytic settings. 0:52:32

Discussion. Asked whether the oldest étale maps are Riemann surfaces, the lecturer agrees, once ramification points are removed, though sheaves were not then in view. 1:01:32

Definition. The notions dual to products, equalizers, fibre products and the terminal object are coproducts, coequalizers, pushouts and the initial object. 1:37:06

Proposition. *In* **Set**, *coproducts are disjoint unions and the initial object is* $\varnothing$. 1:38:08

Proposition. *In* **Set**, *the coequalizer of* $E \rightrightarrows F$ *is the quotient of* F *by the equivalence relation generated by the image of* E *in* $F \times F$. 1:38:08

Example. In groups, the coproduct is the free product; the coproduct of n copies of $\mathbb{Z}$ is the free group on n generators. 1:40:15

Remark. For the usual algebraic structures, limits are computed on the underlying sets, but colimits are not. 1:41:22

Proposition. *If* S *admits limits (resp. colimits) of a given type,* $\mathrm{Hom}(C^{\mathrm{op}}, S)$ *admits them too, computed argumentwise.* 1:43:32

Proposition. *The category opposite to* $\mathrm{Hom}(C^{\mathrm{op}}, S)$ *is* $\mathrm{Hom}(C, S^{\mathrm{op}})$. 1:44:35

Proposition. *Presheaves on* X *admit all colimits, computed open set by open set.* 1:46:41

Counterexample. The subcategory $\mathrm{Sh}(X)$ is not stable in presheaves under any of the usual types of colimit. 1:47:49

Example. The coproduct, taken in presheaves, of the terminal object with itself (the constant presheaf with two elements) is not a sheaf. 1:50:02

Proposition. *For every sheaf* F, $F(\varnothing)$ *is a singleton ($\varnothing$ being covered by the empty family).* 1:52:10

Example. The initial presheaf $U \mapsto \varnothing$ is not a sheaf. 1:53:17

Proposition. *A colimit of sheaves exists if and only if the functor $G \mapsto \mathrm{Hom}(P, G)$ on $\mathrm{Sh}(X)$ is representable, P being the colimit taken in presheaves.* 1:54:21

5 The associated sheaf

Definition. The sheaf associated to a presheaf P is a sheaf $\widetilde{P}$ equipped with $P \to \widetilde{P}$, universal among morphisms from P to sheaves. 1:57:33

Theorem. *Every presheaf admits an associated sheaf, unique up to unique isomorphism.* 1:58:36

He returns to Godement's book for sheaves. [1] 1:59:40

Proposition. $\mathrm{Sh}(X)$ *admits all colimits, the sheaf associated to the colimit computed in presheaves.* 2:01:50

Proposition. *A presheaf and its associated sheaf have the same stalks.* 2:01:50

Proposition. *In terms of étalé spaces, the coproduct of a family of sheaves is the disjoint topological sum, which is étalé over X.* 2:04:10

Proposition. *The coequalizer of $F \rightrightarrows G$ (étalé spaces over X) is G/R, with the quotient topology, R the generated relation; G/R is étalé over X.* 2:05:16

Proposition. *Coproducts and coequalizers of double arrows suffice to construct all colimits.* 2:07:22

6 Exactness properties of sheaves

Remark. Principle: $\mathrm{Sh}(X)$ has the exactness properties of **Set** that involve arbitrary colimits and finite limits. 2:08:24

Counterexample. In **Set**, an arbitrary product of epimorphisms is an epimorphism; this is false in $\mathrm{Sh}(X)$ in general. 2:09:28

Discussion. Infinite products of epimorphisms are epimorphisms in sets, because surjections split there, but not for sheaves; the lecturer warns that infinite inverse limits behave differently and promises to return to it. 2:09:28

Proposition. *Categories of presheaves (of functors to* **Set***) have all the exactness properties of* **Set***, limits and colimits being computed argumentwise.* 2:12:40

Proposition. *The functor $a\colon P \mapsto \widetilde{P}$ is left adjoint to the inclusion i:* $\mathrm{Hom}(P, iF) \simeq \mathrm{Hom}(aP, F)$*; it commutes with colimits.* 2:13:45

Theorem. *The associated-sheaf functor is left exact (it commutes with finite products and equalizers).* 2:14:49

Proposition. *In $\mathrm{Sh}(X)$, coequalizers are stable under base change:* $\mathrm{Coker}(F \rightrightarrows G) \times_H H'$ *is the coequalizer of the inverse images over H'.* 2:18:08

Proposition. *In $\mathrm{Sh}(X)$, coproducts are stable under base change:* $(\coprod_i F_i) \times_F F' \simeq \coprod_i (F_i \times_F F')$. 2:24:32

7 Quotients and equivalence relations

Definition. A quotient object of F is a class of epimorphisms $F \to G$ up to isomorphism under F. 2:27:43

Tape 2, side B

Proposition. *For every morphism $F \to G$, the fibre product $F \times_G F \subset F \times F$ is an equivalence relation on F.* 2:28:43

Proposition. *To an equivalence relation $R \subset F \times F$ is associated the coequalizer of $R \rightrightarrows F$, a quotient of F (an epimorphism).* 2:29:47

Definition. A quotient is effective if it is the quotient by its relation $F \times_G F$; a relation R is effective if $R = F \times_{F/R} F$. 2:30:52

Proposition. *In* $\mathrm{Sh}(X)$*, as in* **Set***, equivalence relations and quotients are effective and correspond bijectively.* 2:31:57

8 Torsors and stalkwise checks

Session of 11 May 1973

Video 1/4 (`URU5jgTvjhU`): tape 1, side A (2:33:29–3:22:55); tape 1, side B (3:22:55–4:02:08); tape 2, side A (4:02:08–4:48:29); tape 2, side B (4:48:29–5:36:41).

Tape 1, side A

Definition. In **Set**, G acts simply transitively on E if $G \times E \to E \times E$, $(g, x) \mapsto (gx, x)$, is bijective; E is a torsor if moreover $E \to \mathrm{pt}$ is surjective. 2:33:33

Proposition. *$E \to F$ is a monomorphism if and only if $E \to E \times_F E$ is an isomorphism; dually, an epimorphism if $F \sqcup_E F \to F$ is an isomorphism.* 2:34:34

Definition. A torsor under a sheaf of groups G is a G-sheaf E with $G \times E \simeq E \times E$ and $E \to 1$ an epimorphism (and not $E(U)$ a torsor for every U). 2:35:38

Proposition. *A morphism of sheaves is an epimorphism if and only if it is one on stalks; E is a G-torsor if and only if each E_x is a G_x-torsor.* 2:36:42

Proposition. *A torsor is trivial (isomorphic to G acting by translations) if and only if it admits a global section; nontrivial ones exist.* 2:38:50

Remark. Principle: statements about structures defined by finite limits and colimits can be checked stalkwise, the fibre functors forming an exact conservative family. 2:40:58

Discussion. The principle that statements proved for sets transfer to sheaves by checking on fibres is recalled with a historical remark: Chevalley devoted a year of his IHÉS seminar around 1958 to formalising it. 2:42:04

Counterexample. The structures of field and of integral domain cannot be expressed by finite limits alone. 2:45:00

Definition. A sheaf of rings K is called integral if $K' \sqcup 1 \to K$ (inclusion and zero section) is an epimorphism, K' being defined below. 2:50:29

Proposition. *For two sheaves K, L, the presheaf $\underline{\mathrm{Hom}}(K, L) \colon U \mapsto \mathrm{Hom}(K|_U, L|_U)$ is a sheaf.* 2:52:38

Proposition. *The sections f of K whose left and right multiplications are monomorphisms of $K|_U$ form a subsheaf K' (a local property).* 2:53:43

9 Generating families and images

Definition. (Z_i) is a generating family of C if every monomorphism $E \to F$ such that $\mathrm{Hom}(Z_i, E) \to \mathrm{Hom}(Z_i, F)$ is bijective for all i is an isomorphism. 2:59:16

Universes, generators and ind-objects are treated in exposé I of SGA 4. [3] 2:59:16

Proposition. *In* $\mathrm{Sh}(X)$*,* (Z_i) *is generating if and only if the family of functors* $\mathrm{Hom}(Z_i, -)$ *is conservative.* 3:01:24

Proposition. *In* $\mathrm{Sh}(X)$*, every morphism factors, essentially uniquely, as an epimorphism followed by a monomorphism (image).* 3:03:48

Proposition. *In* $\mathrm{Sh}(X)$*, a morphism that is both a monomorphism and an epimorphism is an isomorphism.* 3:05:57

Definition. An open set U of X defines a sheaf: $U(V)$ is a singleton if $V \subset U$, empty otherwise. 3:09:09

Proposition. *For every sheaf* E*,* $\mathrm{Hom}(U, E) = E(U)$*; the open sets of* X *therefore form a small generating family of* $\mathrm{Sh}(X)$*.* 3:11:24

10 Abelian sheaves and injectives

Proposition. *The abelian sheaves on* X *(abelian group objects of* $\mathrm{Sh}(X)$*) form an abelian category* $\mathrm{Ab}(X)$*.* 3:12:35

Proposition. $\mathrm{Ab}(X)$ *admits limits and colimits; filtered colimits there commute with the forgetful functor and are exact.* 3:14:47

Definition. An abelian sheaf I is injective if every morphism $F \to I$ extends along every monomorphism $F \to G$. 3:16:56

Proposition. *An abelian category with exact filtered colimits and a small generating family has enough injectives; this is the case of* $\mathrm{Ab}(X)$ *(free abelian sheaves on the* Z_i*).* 3:18:01

Proposition. *For a sheaf of rings* A*,* $\mathrm{Mod}(A)$ *is abelian, with exact filtered colimits and generators, hence has enough injectives (*Ext*, cohomology).* 3:19:05

Counterexample. In general $\mathrm{Ab}(X)$ does not have enough projectives; filtered limits there are not exact. 3:21:12

11 Direct and inverse images

Tape 1, side B

Definition. For $f\colon X \to Y$, an f-morphism $G \to F$ is a family $G(V) \to F(U)$, for $f(U) \subset V$, compatible with the restrictions. 3:24:08

Proposition. *An* f*-morphism is equivalent to a family* $G(V) \to F(f^{-1}(V))$ *compatible with the restrictions.* 3:27:20

Proposition. $V \mapsto f^{-1}(V)$ *is an increasing functor* $\mathcal{O}_Y \to \mathcal{O}_X$ *that commutes with arbitrary unions.* 3:30:33

Definition. Direct image of a presheaf F on X, $f_*F(V) = F(f^{-1}(V))$, a presheaf on Y. 3:31:42

Proposition. *The f-morphisms $G \to F$ correspond bijectively to the morphisms $G \to f_*F$ of presheaves on Y.* 3:32:48

Proposition. *On presheaves, f_* commutes with arbitrary limits and colimits.* 3:33:52

Proposition. *The direct image of a sheaf is a sheaf.* 3:34:59

Proposition. *$f_*\colon \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ commutes with arbitrary limits.* 3:38:17

Counterexample. In general f_* commutes neither with coproducts nor with epimorphisms: for Y a point, f_* is $\Gamma(X, -)$. 3:39:22

Definition. The inverse image f^*G is equipped with a universal f-morphism $G \to f^*G$; f^* is the left adjoint of f_*. 3:41:34

Proposition. *For every functor $\varphi\colon C \to D$, the restriction $\widehat{D} \to \widehat{C}$, $P \mapsto P \circ \varphi$, has a left adjoint and a right adjoint.* 3:44:45

Proposition. *f^*G is the sheaf associated to the presheaf inverse image $f^*_{\mathrm{pr}}G$.* 3:46:54

Proposition. *f^* commutes with arbitrary colimits (left adjoint).* 3:49:03

Theorem. *f^* is left exact (it commutes with finite limits).* 3:50:13

Definition. The inverse image of a space $G \to Y$ under f is $X \times_Y G \subset X \times G$, with the induced topology; it is a fibre product in topological spaces. 3:52:25

Proposition. *The property of being étalé is stable under base change.* 3:53:31

Proposition. *For sheaves viewed as étalé spaces, f^*G corresponds to the fibre product $X \times_Y G$.* 3:54:35

Proposition. *Étalé spaces over Y are stable under finite limits in spaces over Y.* 3:57:48

Proposition. *The fibre of $X \times_Y G$ at x is homeomorphic to the fibre of G at $f(x)$.* 3:57:48

Proposition. *$(f^*G)_x \simeq G_{f(x)}$; hence again the left exactness of f^*.* 3:58:52

Tape 2, side A

Not indexed (sound too degraded to be transcribed).

12 Stalks and epimorphisms

Tape 2, side B

Proposition. *Two morphisms $u, v\colon F \to G$ of sheaves are equal if and only if $u_x = v_x$ for all x: the fibre functors form a faithful family.* 4:48:31

Proposition. *The fibre functors commute with colimits and finite limits and form a conservative family.* 4:50:40

Proposition. *$F \to G$ is an epimorphism of sheaves if and only if every section of G lifts locally to a section of F.* 4:53:05

Counterexample. An epimorphism of sheaves is not in general an epimorphism of presheaves: $\Gamma(F) \to \Gamma(G)$ is not always surjective. 4:55:12

Proposition. *For X compact Hausdorff, $\Gamma(X, -)$ sends epimorphisms of sheaves to surjections if and only if X is totally disconnected.* 4:55:12

13 Algebraic structures on objects of a category

Definition. A composition law on an object E of C is an arrow $E \times E \to E$; commutativity and associativity are expressed by diagrams. 4:59:30

Definition. A left unit is an arrow $1 \to E$ (1 terminal) such that $E \simeq 1 \times E \to E \times E \to E$ is the identity. 5:01:42

Proposition. *Via the Yoneda embedding, a law $E \times E \to E$ is equivalent to laws on the $E(Z) = \mathrm{Hom}(Z, E)$, functorial in Z.* 5:04:55

Definition. An algebraic structure (group, ring, module, etc.) on E is such a structure on each $E(Z)$, functorial in Z. 5:07:07

Proposition. *The structure of a category ($D_1 \rightrightarrows D_0$, composition $D_1 \times_{D_0} D_1 \to D_1$, units) is described by fibre products.* 5:11:27

Definition. A category internal to C is a pair (D_0, D_1) with a category structure on $(D_0(Z), D_1(Z))$, functorial in Z. 5:13:34

Proposition. *The embedding $C \to \widehat{C}$ commutes with arbitrary limits, but not with colimits in general.* 5:14:41

Proposition. *A structure defined by limits on a sheaf F is equivalent to the same structure on each $F(U)$, functorial in U.* 5:15:45

Definition. A sheaf of groups is a functor $\mathcal{O}(X)^{\mathrm{op}} \to$ **Grp** whose composite with the forgetful functor is a sheaf. 5:17:53

Proposition. *The axioms of a structure defined by finite limits can be checked stalkwise.* 5:22:21

Proposition. *A functor that commutes with finite products sends group objects (ring objects, etc.) to objects of the same type.* 5:23:26

Proposition. *The stalks of a sheaf of groups (of rings, etc.) are groups (rings, etc.).* 5:25:32

Definition. For a k-coalgebra A, an A-comodule is a k-module M equipped with $M \to M \otimes_k A$ satisfying coassociativity and counit. 5:27:45

Remark. The tensor product of modules is constructed by colimits (sums, quotients), not by limits alone. 5:30:56

Counterexample. For sheaves of modules, $M(U) \otimes_{k(U)} A(U) \to (M \otimes_k A)(U)$ is in general neither injective nor surjective. 5:33:19

Definition. E is a torsor under a group G if $E \neq \varnothing$ and G acts simply transitively on E. 5:34:24

14 Étale morphisms and slice categories

Session of 18 May 1973

Video 1/4 (URU5jgTvjhU): tape 1, side A (5:36:41–6:26:00); tape 1, side B (6:26:00–7:14:32); tape 2, side A (7:14:32–8:03:53); tape 2, side B (8:03:53–8:23:34).

Tape 1, side A

Proposition. *For a point y of Y and the inclusion $i_y : \{y\} \to Y$, we have $i_y^* F \simeq F_y$: the stalk is the inverse image under a point.* 5:39:20

Proposition. *If $f : X \to Y$ is étale, X is an object of $\mathrm{Sh}(Y)$ and $\mathrm{Sh}(X) \simeq \mathrm{Sh}(Y)/X$; a composite of étale morphisms is étale.* 5:41:31

Definition. For an object X of a category C, the category C/X has as objects the arrows $X' \to X$ and as morphisms the commutative triangles. 5:43:40

Proposition. *If $f : X \to Y$ is étale, f^*G is the product $G \times X$ in $\mathrm{Sh}(Y)$, viewed as an object of $\mathrm{Sh}(Y)/X$.* 5:44:44

Proposition. *For $g : X' \to X$ in $\mathrm{Sh}(Y)$, the functor $g^* : \mathrm{Sh}(Y)/X \to \mathrm{Sh}(Y)/X'$ is the base change $F \mapsto F \times_X X'$.* 5:47:00

Proposition. *f_* sends every structure defined by limits (group, ring, etc.) to a structure of the same kind, but not torsors.* 5:51:15

Proposition. *f^* respects structures defined by finite limits and colimits: the inverse image of a G-torsor is an f^*G-torsor.* 5:52:19

15 Ring and field objects

Definition. A ring object A of C is an object equipped with a ring structure on each $A(Z) = \mathrm{Hom}_C(Z, A)$, functorial in Z. 5:56:33

Proposition. *The units $A^*(Z) \subset A(Z)$ form a group-valued functor, represented by the fibre product of $m : A \times A \to A$ and $1 : e \to A$.* 5:57:38

Definition. A ring object A is a field object if the morphism $A^* \sqcup e \to A$ (inclusion and zero section) is an isomorphism. 5:59:47

Definition. Integrality, first condition: for $C = m^{-1}(0) \subset A \times A$, the morphism $A \sqcup A \to C$, $x \mapsto (x, 0)$, $x \mapsto (0, x)$, is an epimorphism. 6:02:58

Proposition. *A morphism $S \to C$ is an epimorphism iff the canonical arrow $C \sqcup_S C \to C$ is an isomorphism.* 6:07:15

Definition. Integrality, second condition ($0 \neq 1$): the kernel $e' \subset e$ of $0, 1 : e \rightrightarrows A$ is initial; in $\mathrm{Sh}(X)$, e' is the largest open set where $A = 0$. 6:08:19

Proposition. *Every kind of structure defined by finite limits and arbitrary colimits admits a "classifying topos", equipped with a universal model.* 6:11:32

Remark. Normal rings, reduced rings, rings with connected spectrum can also be defined by such diagrammatic axioms. 6:12:37

Proposition. *A sheaf of rings is never a presheaf of fields: its value on $\varnothing$ is the zero ring.* 6:13:40

Discussion. A question shows that a sheaf of fields in the usual sense can never be a field object: sections over disjoint opens form a product of fields, and the empty open gives the zero ring. 6:13:40

Proposition. *On a space, such an axiom is checked stalkwise: a sheaf of rings A is a field object iff each stalk A_x is a field.* 6:15:45

16 Constant sheaves and sheaves of fields

Definition. The constant sheaf $I_X = p^*I$ ($p : X \to \mathrm{pt}$) has étalé space $I \times X$; it is the coproduct of I copies of the terminal object. 6:16:49

Proposition. *If I is a group, a ring or a field, I_X is a sheaf of groups, of rings or of fields; for example K_X for a field K.* 6:19:00

Example. On a reduced and locally irreducible complex analytic space, the sheaf $\mathcal{M}_X$ of meromorphic functions is a sheaf of fields. 6:20:05

Proposition. *The stalk $\mathcal{M}_{X,x}$ is the total ring of fractions of $\mathcal{O}_{X,x}$; it is a field as soon as $\mathcal{O}_{X,x}$ is a domain.* 6:21:07

Proposition. *On an integral scheme, the sheaf of rational functions is a constant sheaf of fields; the analytic sheaf $\mathcal{M}_X$ is not.* 6:22:14

17 Internal Hom and the subobject sheaf

Definition. $\mathcal{H}om(F,G)(U) = \mathrm{Hom}(F|_U, G|_U)$ defines a sheaf on X, to be distinguished from the set $\mathrm{Hom}(F,G)$. 6:23:18

Tape 1, side B

Proposition. *For X' étale over X:* 6:26:09
$\mathrm{Hom}(X', \mathcal{H}om(F,G)) \simeq \mathrm{Hom}_{X'}(F \times X', G \times X') \simeq \mathrm{Hom}(F \times X', G)$.

Definition. In a category with products, $\mathcal{H}om(F,G)$ is, if it exists, the object representing $X' \mapsto \mathrm{Hom}(F \times X', G)$; it exists in $\mathrm{Sh}(X)$. 6:29:26

Definition. For group objects F, G, $\mathcal{H}om_{\mathrm{gr}}(F,G)$ represents $X' \mapsto \mathrm{Hom}_{\mathrm{gr}}(F_{X'}, G_{X'})$, homomorphisms of group objects of C/X'. 6:30:28

Definition. The sheaf $\mathcal{P}(F)$: $\mathcal{P}(F)(U)$ is the set of subsheaves of $F|_U$; for X a point, it is the set of subsets of F. 6:32:35

Proposition. *$\mathcal{P}(F)$ represents $X' \mapsto \mathrm{Sub}(F \times X')$; the subobjects of an object of C/X' are the same in C and in C/X'.* 6:34:44

Proposition. *For the terminal object e, $\mathcal{P}(e)$ is the sheaf that associates to U the set of its open subsets; it represents $X' \mapsto \mathrm{Sub}(X')$.* 6:38:01

Proposition. *If $\mathcal{P}(e)$ exists, $\mathcal{P}(F)$ exists iff $\mathcal{H}om(F, \mathcal{P}(e))$ exists, and they are isomorphic; in* **Set**, *$\mathcal{P}(e)$ has two elements.* 6:41:10

Proposition. *A subsheaf of $\mathcal{P}(F)$ is a family of sets of subsheaves of the $F|_U$, stable under restriction and local in U.* 6:44:25

Definition. A topology on a sheaf F is defined as a subsheaf of $\mathcal{P}(F)$ satisfying the usual axioms, read in $\mathrm{Sh}(X)$. 6:46:35

Proposition. *For $f : X \to Y$, there are canonical arrows $f^*\mathcal{H}om(F,G) \to \mathcal{H}om(f^*F, f^*G)$ and $f^*\mathcal{P}(F) \to \mathcal{P}(f^*F)$.* 6:47:39

Counterexample. These arrows are not isomorphisms in general, for example for a fibre functor; they are if f is étale. 6:50:49

18 Stalks and infinite products

Proposition. *f^* does not commute with infinite products in general; there is a canonical arrow $(\prod_i F_i)_x \to \prod_i (F_i)_x$.* 6:51:54

Remark. Its injectivity means: if f_i, g_i have the same germ at x for every i, they coincide on a common neighbourhood of x. 6:54:11

Counterexample. U_i open neighbourhoods of x whose intersection is not a neighbourhood, F the sheaf of maps to $\{a, b\}$, $f_i = a$, $g_i = a$ on U_i and b elsewhere. 6:55:19

Proposition. *The stalk functor at x commutes with products iff x has a smallest open neighbourhood; if points are closed, iff x is isolated.* 6:57:25

Counterexample. In $\mathrm{Sh}(X)$, an infinite product of epimorphisms is not always an epimorphism; one reduces to the case of the $F_i \to e$. 6:58:33

Example. X locally connected, x closed, U_i open neighbourhoods of x whose intersection is not a neighbourhood, $F_i = U_i \sqcup (X \setminus \{x\})$; $\prod_i F_i \to e$ is not an epimorphism. 7:01:47

Discussion. Looking for a counterexample to products of epimorphisms of sheaves being epimorphisms, the lecturer finds one on locally connected spaces such as a line, at any non-isolated point, but none on totally disconnected compact ones. 7:01:47

Proposition. *If X is compact Hausdorff and totally disconnected, $F \to G$ is an epimorphism iff $F(U) \to G(U)$ is surjective for every clopen set U; every product of epimorphisms is an epimorphism.* 7:05:03

19 Connectedness and cohomology

Proposition. *X is connected iff the terminal object of $\mathrm{Sh}(X)$ is not a sum of two non-initial subobjects.* 7:12:57

Tape 2, side A

Proposition. *X is nonempty iff $\varnothing \to e$ is not an isomorphism in $\mathrm{Sh}(X)$; X is empty iff $\mathrm{Sh}(X)$ is equivalent to the point category.* 7:14:35

Definition. The abelian group objects of $\mathrm{Sh}(X)$ form a category with enough injectives; $H^i(X, F) = R^i\Gamma(F)$, with $\Gamma(F) = \mathrm{Hom}(e, F)$. 7:16:42

Cohomology of topoi is in exposé V of SGA 4. [3] 7:16:42

Definition. For an abelian group M, we set $H^i(X, M) = H^i(X, M_X)$; these groups depend only on the category $\mathrm{Sh}(X)$. 7:18:48

Proposition. *X is connected iff $\mathbb{Z} \to H^0(X, \mathbb{Z})$ is surjective; connected and nonempty iff it is bijective.* 7:19:52

Proposition. *The sections of I_X over U are the locally constant maps $U \to I$.* 7:20:57

Remark. The lattice of open sets of X is that of the subobjects of the terminal object of $\mathrm{Sh}(X)$; the Čech constructions (Artin–Mazur) follow from it. 7:23:07

20 Covering spaces and torsors

Homotopy invariants of a topos are the subject of Artin and Mazur's book. [4] 7:23:07

Theorem. *If X is connected, locally connected and locally simply connected, the covering spaces of X correspond to $\pi_1(X, x)$-sets.* 7:24:10

Proposition. *The covering spaces of X are the locally constant objects X' of* $\mathrm{Sh}(X)$*:* $X' \times U_i \simeq \coprod_{S_i} U_i$ *for a covering* $e = \sup_i U_i$. 7:25:16

Proposition. *For X connected and locally connected but not locally simply connected, the covering spaces are classified by a pro-group.* 7:26:21

Proposition. *For a sheaf of groups G not necessarily commutative, $H^1(X, G)$ is the set of isomorphism classes of G-torsors.* 7:27:24

Definition. A G-torsor is an object P equipped with an action of G such that $G \times P \to P \times P$ is an isomorphism and $P \to e$ an epimorphism. 7:28:31

Proposition. $G \mapsto G_X = \coprod_G e$ *is left exact; the G_X-torsors are the Galois covering spaces with group G.* 7:29:35

Proposition. *For X connected, locally connected and locally simply connected, $H^1(X, G)$ is identified with* $\mathrm{Hom}(\pi_1(X, x), G)$ *modulo conjugation; X is simply connected iff all these H^1 are trivial.* 7:30:40

21 Stacks and gerbes

Definition. A fibred category over $\mathcal{O}(X)$ associates to U a category F_U, and to $V \subset U$ a restriction functor, transitive up to coherent isomorphism. 7:31:44

Example (which are stacks). $U \mapsto \mathrm{Sh}(U)$, fibre spaces over U, sheaves of groups on U, $G|_U$-torsors. 7:32:47

Definition. A fibred category is a stack if the presheaves $\mathcal{H}om(F, G)$ are sheaves and if every gluing datum (cocycle on the U_{ijk}) is effective. 7:34:51

Definition. For G noncommutative, $H^2(X, G)$ is defined as a set of classes of stacks (gerbes) attached to G (Giraud); it depends only on $\mathrm{Sh}(X)$. 7:39:09

Non-abelian cohomology and stacks are in Giraud's book. [5] 7:39:09

Definition. A stack over $\mathcal{E}$ is a fibred category over $\mathcal{E}$ with descent along epimorphic families, the intersections becoming fibre products. 7:44:33

Proposition. *Restriction and extension give an equivalence between stacks over $\mathcal{O}(X)$ and stacks over* $\mathrm{Sh}(X)$. 7:45:38

22 Gluing and quotient topoi

Proposition. $\mathrm{Sh}(\coprod_i X_i) \simeq \prod_i \mathrm{Sh}(X_i)$*: the sum of spaces corresponds to the product of categories.* 7:48:59

Proposition. *If Z is closed in X and in Y,* $\mathrm{Sh}(X \cup_Z Y)$ *is the 2-fibre product of* $\mathrm{Sh}(X)$ *and* $\mathrm{Sh}(Y)$ *over* $\mathrm{Sh}(Z)$. 7:50:03

Proposition. *For an equivalence relation R on X, the sheaves on X equipped with descent data relative to R form a quotient topos, even if X/R is pathological.* 7:52:11

23 Irreducible closed sets and sober spaces

Definition. A space Z is irreducible if it is nonempty and not a union of two closed subsets of Z distinct from Z. 7:54:19

Proposition. *A separated space is irreducible iff it is a point; a variety is irreducible for the Zariski topology iff it is so in the algebraic sense.* 7:55:24

Proposition. *The closure of an irreducible subset is irreducible.* 7:55:24

Definition. We write $\tilde{X}$ for the set of irreducible closed subsets of X and $i : X \to \tilde{X}$, $x \mapsto \overline{\{x\}}$; i is bijective if X is separated. 7:56:30

Proposition. *The subsets $\tilde{U} = \{Y \in \tilde{X} : Y \cap U \neq \varnothing\}$, U open, form a topology: $U \mapsto \tilde{U}$ respects unions and finite intersections.* 7:57:34

Proposition. *We have $i^{-1}(\tilde{U}) = U$; i is continuous and induces a lattice isomorphism $\mathcal{O}(\tilde{X}) \simeq \mathcal{O}(X)$.* 8:00:50

Proposition. *Consequently $i_* : \mathrm{Sh}(X) \to \mathrm{Sh}(\tilde{X})$ is an isomorphism of categories, with inverse i^*.* 8:03:02

Tape 2, side B

Proposition. *$i : X \to \tilde{X}$ is injective iff X is T_0; if X is separated, i is a homeomorphism.* 8:03:56

Definition. A generic point of a closed set Z is a point whose closure is Z; i is surjective iff every irreducible closed set has a generic point. 8:05:01

Example. A variety over K reduced to its K-rational points has irreducible closed sets without a generic point. 8:06:06

Definition. X is sober if every irreducible closed set has exactly one generic point, i.e. if $i : X \to \tilde{X}$ is bijective, hence a homeomorphism. 8:07:14

Definition. For Y an irreducible closed set, the stalk $F_Y = \varinjlim_{U \cap Y \neq \varnothing} F(U)$ (filtered colimit) is left exact and commutes with colimits. 8:08:20

Proposition. *$F_{\overline{\{x\}}} = F_x$; in general F_Y is the stalk at the point Y of F viewed as a sheaf on $\tilde{X}$.* 8:10:27

Theorem. *$Y \mapsto (F \mapsto F_Y)$ is a bijection from $\tilde{X}$ onto the classes of fibre functors $\mathrm{Sh}(X) \to$ **Set** (left exact, commuting with $\varinjlim$).* 8:11:30

Proposition. *A fibre functor on $\mathrm{Sh}(X)$ has no nontrivial automorphism.* 8:12:39

Theorem. *$\tilde{X}$ is sober, and $X \mapsto \tilde{X}$ is left adjoint to the inclusion of sober spaces: $\mathrm{Hom}(X, Y) \simeq \mathrm{Hom}(\tilde{X}, Y)$ for Y sober.* 8:13:42

Definition. On the set $\Phi(\mathcal{E})$ of classes of fibre functors, the sets $\tilde{U} = \{\varepsilon : \varepsilon(U) \neq \varnothing\}$, $U \subset e$, form a topology. 8:16:56

The exercises of SGA 4 are recommended. [3] 8:16:56

Theorem. *$X \to \Phi(\mathrm{Sh}(X))$ is a homeomorphism iff X is sober: a sober space can be reconstructed from $\mathrm{Sh}(X)$.* 8:20:13

Proposition. *If X is T_1 (for example the closed points of a Jacobson scheme), X is identified with the subspace of closed points of $\tilde{X}$.* 8:21:16

24 Points and morphisms of sober spaces

Session of 25 May 1973

Video 1/4 (URU5jgTvjhU): tape 1, side A (8:23:34–9:28:43); tape 1, side B (9:28:43–10:33:42); tape 2, side A (10:33:42–11:06:44).

Tape 1, side A

Proposition. *For $f : X \to Y$ continuous, the point $f(x)$ corresponds to the fibre functor $\varepsilon_x \circ f^*$, since $(f^*F)_x \simeq F_{f(x)}$.* 8:24:40

Theorem. *If X and Y are sober, every functor $\mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ that is left exact and commutes with $\varinjlim$ is isomorphic to f^* for a unique continuous f.* 8:27:51

Proposition. *Two functors on $\mathrm{Sh}(Y)$ that commute with colimits and are isomorphic on the generating subcategory $\mathcal{O}(Y)$ are isomorphic.* 8:29:59

Definition. For $f, g : X \to Y$, a morphism from f to g is a morphism between f^* and g^*; if Y is separated, there is one only if $f = g$, and it is the identity. 8:33:13

25 Definitions of a topos

Definition (first form). A topos is a category having the exactness properties of **Set** (arbitrary $\varinjlim$, finite $\varprojlim$) and a small generating set. This first form is provisional: it is made precise by the axioms given a few minutes later. 8:37:46

Definition. $C \subset \mathrm{Ob}\,\mathcal{E}$ is generating if $\mathcal{E} \to \widehat{C}$, $X \mapsto \mathrm{Hom}(-, X)|_C$, is faithful. 8:38:52

Proposition. *In a topos, it amounts to the same whether this functor is faithful, fully faithful, conservative, or conservative on monomorphisms.* 8:40:58

Definition (Giraud). A topos is a category with (a) finite $\varprojlim$, (b) universal disjoint coproducts, (c) universal effective equivalence relations, (d) a small generator. 8:44:06

Definition. A coproduct $X = \coprod X_i$ is universal if for every $X' \to X$, the arrow $\coprod_i (X_i \times_X X') \to X'$ is an isomorphism. 8:46:13

Definition. It is disjoint if the $X_i \to X$ are monomorphisms and $X_i \times_X X_j$ is initial for $i \neq j$. 8:48:24

Definition. An equivalence relation $R \subset X \times X$ (each $R(Z)$ being one on $X(Z)$) is effective if $R \simeq X \times_{X/R} X$. 8:51:36

Definition. It is universal if $X \to X/R$ is a universal epimorphism and if passage to the quotient commutes with base changes $Y' \to X/R$. 8:52:42

Proposition. *In a topos, the subobjects and the quotient objects of an object form small sets.* 8:55:52

26 Cohomology in a topos

Proposition. *$\mathcal{E}_{ab}$ is abelian, with exact filtered colimits and a generator; it therefore has enough injectives.* 8:56:57

Proposition. *An injective object of $\mathcal{E}_{ab}$ is injective in $\mathcal{E}$; in particular $\mathcal{E}$ has enough injectives.* 8:58:03

Definition. In a topos, $\Gamma(X) = \mathrm{Hom}(e, X)$; its derived functors on $\mathcal{E}_{ab}$ are the $H^i(\mathcal{E}, -)$. 9:02:21

Definition. The constant object $I_{\mathcal{E}} = \coprod_I e$: $I \mapsto I_{\mathcal{E}}$ commutes with $\varinjlim$ and finite $\varprojlim$; we set $H^i(\mathcal{E}, M) = H^i(\mathcal{E}, M_{\mathcal{E}})$. 9:04:28

Remark. A topos has homotopy invariants (pro-groups, the Čech method of Artin–Mazur), which recover the π_i of a locally contractible space. 9:06:45

27 Basic examples of topoi

Artin and Mazur again, for homotopy invariants. [4] 9:06:45

Again the exercises of SGA 4. [3] 9:08:54

Proposition. $\mathrm{Sh}(\varnothing)$ *is equivalent to the point category: it is the "empty topos".* 9:09:57

Proposition. *A topos $\mathcal{E}$ is empty iff the arrow $\varnothing_{\mathcal{E}} \to e_{\mathcal{E}}$ is an isomorphism.* 9:12:01

Proposition. $\mathrm{Sh}(\mathrm{pt}) \simeq$ **Set** *via $F \mapsto F(\mathrm{pt})$: it is the point topos, or "final topos".* 9:14:08

Proposition. *A topos $\mathcal{E}$ is a point iff the functor* **Set** $\to \mathcal{E}$, $I \mapsto I_{\mathcal{E}}$, *is an equivalence.* 9:17:23

Proposition. *For a discrete group G acting on X, the category $\mathrm{Sh}(X, G)$ of étalé spaces over X equipped with a compatible G-action is a topos.* 9:18:25

Definition. $\mathrm{Sh}(X, \{1\}) = \mathrm{Sh}(X)$; $\mathrm{Sh}(\mathrm{pt}, G) = B_G$, the category of G-sets, is the "classifying topos" of G. 9:21:38

Proposition. $(B_G)_{ab}$ *is the category of G-modules, $\Gamma(M) = M^G$ and $H^i(B_G, M) = H^i(G, M)$, the cohomology of the classifying space of G.* 9:23:42

Proposition. *The cohomology of $\mathrm{Sh}(X, G)$ is the equivariant cohomology $H^i(X; G, M)$ (Tôhoku).* 9:25:54

28 Points of a topos

Tape 1, side B

Cohomology with operators $H^i(X; G, M)$ is in his 1957 paper on homological algebra. [6] 9:27:02

Counterexample. If $G \neq 1$, the topos B_G is equivalent to no $\mathrm{Sh}(X)$. 9:28:50

Example. The forgetful functor $B_G \to$ **Set** is a fibre functor, with automorphism group G. 9:30:01

Proposition. *If X is sober, every fibre functor on $\mathrm{Sh}(X)$ is isomorphic to $F \mapsto F_x$ for a unique $x \in X$.* 9:32:08

Proposition. *Specialization ($y \in \overline{\{x\}}$) is a preorder, which is an order if X is sober; it induces $F_y \to F_x$, functorial in F.* 9:33:13

Proposition. *There is a morphism $\varepsilon_y \to \varepsilon_x$ iff y is a specialization of x, and it is unique: the points of $\mathrm{Sh}(X)$ form an ordered set.* 9:37:42

Example. B_G has a single point up to isomorphism without being the final topos: its points have nontrivial automorphisms. 9:38:48

Proposition. *In $\mathrm{Sh}(X, G)$, $\varepsilon_x^G : F \mapsto F_x$ is a point, with automorphism group the stabilizer G_x.* 9:40:58

29 Giraud's theorem and elementary topoi

Theorem (Giraud). *$\mathcal{E}$ is a topos iff there exist C small and $i : \mathcal{E} \to \widehat{C}$ fully faithful with a left exact left adjoint.* 9:46:33

Proposition. *For every generating full subcategory C of a topos, $\mathcal{E} \to \widehat{C}$ is fully faithful and has a left exact left adjoint.* 9:51:03

Proposition. *Topoi in the sense of Giraud's axioms are exactly the categories of sheaves on a site.* 9:53:11

Proposition. *In a topos, $\mathcal{H}om(X, Y)$ exists: it represents $Z \mapsto \mathrm{Hom}(X \times Z, Y)$.* 9:55:24

Proposition. *In a topos, the functor $Z \mapsto \mathrm{Sub}(Z)$ is representable by an object $P_{\mathcal{E}}$.* 9:56:29

Proposition. *In a topos, $P : \mathcal{E}^\circ \to$ **Set** is representable iff it sends colimits to limits.* 9:57:40

Accessible functors are in exposé I, § 9 of SGA 4, which he says tells the matter somewhat inadequately. [3] 9:59:51

Proposition. *Every topos is an elementary topos (Lawvere–Tierney: finite $\varprojlim$, $\mathcal{H}om$, $P_{\mathcal{E}}$); an elementary topos with small coproducts and a small generator is a topos.* 10:00:55

30 Presheaf topoi and the classifying topos of a group

Discussion. Lawvere's elementary topoi are compared with Giraud's: more general, with finitary axioms, though internal Hom is a very strong operation; adding generators and limits recovers sheaf topoi. 10:03:02

Proposition. *In $\mathrm{Sh}(X, G)$, limits, $\mathcal{H}om$ and P are computed in $\mathrm{Sh}(X)$, with the induced action of G.* 10:05:07

Proposition. *For C small, $\widehat{C}$ is a topos (limits argumentwise, representables generating), with all the exactness properties of* **Set**. 10:07:16

Example. $B_G = \widehat{C}$ for C the category with one object whose endomorphisms are G. 10:10:30

Proposition. *For $X \in C$, $F \mapsto F(X)$ is a point of $\widehat{C}$; every point is a filtered colimit of these, and $\widehat{C}$ has enough points.* 10:11:33

Proposition. *$\mathrm{Sh}(X, G)$ has enough points: the ε_x^G form a conservative family.* 10:12:36

Definition. For a group object G of a topos $\mathcal{E}$, the objects of $\mathcal{E}$ equipped with an action of G form a topos B_G; for $\mathcal{E} =$ **Set**, the G-sets. 10:14:42

Proposition. *$H^i(G, M) = H^i(B_G, M)$ for M abelian with an action of G; $\Gamma(X)$ is the set of G-invariant sections of X.* 10:17:55

Theorem. *B_G classifies G-torsors: for $f : \mathcal{T} \to \mathcal{E}$, the morphisms $\mathcal{T} \to B_G$ over $\mathcal{E}$ correspond to the f^*G-torsors.* 10:21:11

31 Induced topoi

Proposition. *For an object X of a topos $\mathcal{E}$, $\mathcal{E}/X$ is a topos, the induced topos; $\mathrm{Sh}(Y)/X \simeq \mathrm{Sh}(X)$ for X étale over Y.* 10:23:20

32 Morphisms of topoi

Proposition. *Between topoi, every functor f^* commuting with colimits has a right adjoint f_* (representability lemma).* 10:32:10

Tape 2, side A

Definition. A morphism of topoi $f : \mathcal{E} \to \mathcal{F}$ is a pair of adjoint functors $f^* \dashv f_*$, with $f^* : \mathcal{F} \to \mathcal{E}$ left exact. 10:35:54

Definition. The morphisms $\mathcal{E} \to \mathcal{F}$ form a category $\mathbf{Homtop}(\mathcal{E}, \mathcal{F})$, whose arrows $f \to g$ are the morphisms $f_* \to g_*$ (or $g^* \to f^*$). 10:38:05

Proposition. *Composition: $(gf)_* = g_* f_*$ and $(gf)^* = f^* g^*$; it is associative.* 10:45:28

Proposition. *Composition is a functor $\mathbf{Homtop}(\mathcal{E}, \mathcal{F}) \times \mathbf{Homtop}(\mathcal{F}, \mathcal{G}) \to \mathbf{Homtop}(\mathcal{E}, \mathcal{G})$: topoi form a 2-category.* 10:47:39

Proposition. *A morphism f is an equivalence of topoi iff f^*, or f_*, is an equivalence of categories.* 10:53:57

Proposition. *For Y sober, the category of points $\mathbf{Homtop}(\mathrm{pt}, \mathrm{Sh}(Y))$ is equivalent to Y ordered by specialization.* 10:59:20

Proposition. *For X, Y sober, $\mathbf{Homtop}(\mathrm{Sh}(X), \mathrm{Sh}(Y))$ is equivalent to the continuous maps f ordered by $f \leq g$ iff $g(x) \in \overline{\{f(x)\}}$ for all x.* 11:01:29

Proposition. *If Y is separated, this category is discrete.* 11:04:45

33 Points of a topos and specialization

Session of 30 May 1973

Video 2/4 (PqCQM3Pcl6Q): tape 1, side A (0:00:00–0:49:07); tape 1, side B (0:49:07–1:38:30); tape 2, side A (1:38:30–1:55:37).

Tape 1, side A

Remark. A morphism of topoi $f\colon E \to F$ is a pair $f^*\colon F \to E$, $f_*\colon E \to F$ with $f^* \dashv f_*$ and f^* left exact; topoi, their morphisms and the natural transformations between morphisms form a 2-category, with hom-categories $\mathrm{Hom}_{\mathrm{top}}(E, F)$ (recalled). 0:04:34

Proposition. *For sober spaces X and Y, the morphisms of topoi $\mathrm{Top}(X) \to \mathrm{Top}(Y)$ come, uniquely, from continuous maps $X \to Y$ (recalled).* 0:10:40

Definition. A point of a topos E is a morphism of topoi $\mathbf{Set} \to E$; the category of points is opposite to the category of fibre functors on E (the inverse images of the points). 0:12:45

Proposition. *For y in a space Y, the fibre functor $F \mapsto F_y = \varinjlim_{U \ni y} F(U) = \mathrm{Hom}(\text{“}\varprojlim\text{”} U, F)$ is represented by the pro-object of $\mathcal{O}_Y$ given by the open neighbourhoods of y.* 0:14:46

Proposition. *Fibre functors on $\mathrm{Top}(Y)$ are represented by pro-objects of $\mathcal{O}_Y$ (by objects exactly for the points having a smallest open neighbourhood, such as the open points), and morphisms between fibre functors correspond to morphisms of the representing pro-objects in the opposite direction.* 0:16:51

Theorem. *For y, y' in a space Y, $\mathrm{Hom}(\varepsilon_y, \varepsilon_{y'})$ has at most one element, and is non-empty exactly when $y \in \overline{\{y'\}}$. So, for Y sober, the category of fibre functors is the set Y ordered by specialization, and the category of points is its opposite.* 0:24:05

Proposition. *For continuous maps $f, g\colon X \to Y$, the set of 2-morphisms $f \Rightarrow g$ between the associated morphisms of topoi has at most one element, and is non-empty exactly when $g(x) \in \overline{\{f(x)\}}$ for every $x \in X$ (up to the convention on the direction of arrows).* 0:27:12

Proposition. *For X sober, every fibre functor on $\mathrm{Top}(X)$ is the stalk at a point of X (an exercise, treated in Hakim's book).* 0:35:00

Proposition. *Hence $X \mapsto \mathrm{Top}(X)$ is a 2-functor from spaces to topoi, the category $\underline{\mathrm{Hom}}(X, Y)$ being the set of continuous maps $X \to Y$ ordered by pointwise specialization (taking sobrifications if needed).* 0:37:03

34 Topoi of spaces with a group action

Definition. For a discrete group G acting on a space X, $\mathrm{Top}(X, G)$ is the topos of G-equivariant sheaves on X (étale spaces over X with a compatible G-action); it is functorial in the pair (X, G) for equivariant maps over group homomorphisms. 0:39:21

Proposition. *For an equivariant map $(X, G) \to (X', G')$, the inverse image of a G'-equivariant étale space $Y' \to X'$ is $Y' \times_{X'} X$ with its unique compatible G-action; this functor commutes with colimits and finite limits, hence defines a morphism of topoi $\mathrm{Top}(X, G) \to \mathrm{Top}(X', G')$.* 0:40:25

Counterexample. The functor $(X, G) \mapsto \mathrm{Top}(X, G)$ is not fully faithful, and (X, G) is not recovered from its topos in general. 0:43:35

Proposition. *If $H \lhd G$ is a normal subgroup acting freely and properly discontinuously on X, then $\mathrm{Top}(X/H, G/H) \simeq \mathrm{Top}(X, G)$.* 0:43:35

Remark. Quotient topoi of this kind are the natural framework for moduli problems (abelian varieties, curves of genus g), where no space with the right properties exists. 0:45:41

Proposition. *For G trivial, $\mathrm{Top}(X, 1) = \mathrm{Top}(X)$; there is a morphism $\mathrm{Top}(X) \to \mathrm{Top}(X, G)$ whose inverse image forgets the G-action, but in general none in the other direction unless G acts trivially.* 0:45:41

Tape 1, side B

35 Points of the equivariant topos

Proposition. *For $x \in X$, the functor sending a G-equivariant sheaf to the stalk at x of its underlying sheaf is a fibre functor on $\mathrm{Top}(X, G)$; these functors are jointly conservative.* 0:49:07

Proposition. *For X sober, the points x, x' of $\mathrm{Top}(X, G)$ are isomorphic iff they lie in the same G-orbit, and $\mathrm{Isom}(x, x') = \{g \in G : gx = x'\}$; the set of isomorphism classes of points is X/G.* 0:50:15

Remark. The topology on the set of isomorphism classes of points is the quotient topology of X/G; if G acts freely and properly, $\mathrm{Top}(X, G)$ is the topos of X/G. 0:52:16

36 The classifying topos of a group

Definition. For a group G, the classifying topos $BG = \mathrm{Top}(\mathrm{pt}, G)$ is the topos of G-sets; a homomorphism $G \to G'$ gives a morphism $BG \to BG'$. 0:53:20

37 Presheaf topoi and the functors u^*, $u_!$, u_*

Definition. For a small category C, $\widehat{C}$ is the topos of presheaves on C (contravariant functors $C \to$ **Set**); BG is the case of a one-object category. 0:55:28

Proposition. *In $\widehat{C}$, $\Gamma(F) = \mathrm{Hom}(1, F) = \varprojlim F$; for F with values in abelian groups, $H^i(\widehat{C}, F) = \varprojlim{}^i F$, the derived functors of the inverse limit.* 0:55:28

Proposition. *A functor $u\colon C \to C'$ induces $u^*\colon \widehat{C'} \to \widehat{C}$, $F' \mapsto F' \circ u$, which commutes with all limits and colimits; it is the inverse image of a morphism of topoi $\widehat{C} \to \widehat{C'}$, whose direct image u_* is the right adjoint of u^*.* 0:58:40

Proposition. *For $u\colon C \to C'$, u^* has both a left adjoint $u_!$ and a right adjoint u_* (Kan extensions); more generally, the inverse image f^* of a morphism of topoi has a left adjoint $f_!$ exactly when it commutes with arbitrary limits (granting a smallness condition left unstated in the lecture).* 1:02:50

Example. For $C' = \Delta$ and $C = \Delta_{\leq n}$ (finite non-empty ordinals of cardinal $\leq n+1$), u^* is truncation of simplicial sets, $u_!$ is the skeleton and u_* the coskeleton. 1:02:50

Definition. When they exist, $f_!$ denotes the left adjoint of f^*, and $f^!$ the right adjoint of f_* (which exists when f_* commutes with all colimits). 1:07:07

Proposition. *If u^* has a left adjoint $u_!$ which is left exact, then $u_!$ is the inverse image of a morphism of topoi in the opposite direction, of which u^* is the direct image.* 1:08:12

Proposition. *If C has finite limits and $u\colon C \to C'$ preserves them, then $u_!$ is left exact, hence the inverse image of a morphism of topoi $\widehat{C'} \to \widehat{C}$.* 1:10:17

Proposition. *Without exactness hypotheses on u, a functor $C \to C'$ gives only a morphism $\widehat{C} \to \widehat{C'}$ in the same direction, and $C \mapsto \widehat{C}$ is a 2-functor (natural transformations give 2-morphisms).* 1:12:22

Proposition. *The functor $\mathrm{Hom}(C, C') \to \mathrm{Hom}_{\mathrm{top}}(\widehat{C}, \widehat{C'})$ is fully faithful; when C, C' are idempotent complete, the morphisms whose inverse image also commutes with all limits are exactly those coming from functors $C \to C'$.* 1:13:27

38 Final and initial topoi

Proposition. *For every topos E there is, up to unique isomorphism, exactly one morphism $E \to$ **Set**, with inverse image $I \mapsto I \cdot 1 = \coprod_I 1$; the one-point topos **Set** is final among topoi.* 1:15:41

Proposition. *The direct image of this unique morphism is the global-section functor $\Gamma = \mathrm{Hom}(1, -)$.* 1:19:55

Proposition. *The inverse image **Set** $\to E$ commutes with infinite products iff E is locally connected (every object is a sum of connected objects); this expresses that products distribute over arbitrary sums in E.* 1:24:13

Question. Are the topoi of the form $\widehat{C}$ exactly those that are locally connected and, in a suitable sense, totally disconnected, i.e. those sharing all exactness properties of **Set**, including those involving infinite limits? 1:26:22

Proposition. *The empty topos* $\mathrm{Top}(\varnothing)$ *(the one-point category) is initial among topoi: each topos receives a unique morphism from it.* 1:28:35

Proposition. *The empty topos is strictly initial: a topos admitting a morphism to it is itself empty.* 1:31:41

Counterexample. Strict initiality of the initial object fails in groups: the trivial group is initial and final, yet non-trivial groups map to it. 1:33:48

39 Classifying topoi in a topos, and sums

Definition. For a group object G of a topos E, the classifying topos $B_E G$ is the topos of objects of E with a G-action; a homomorphism $G \to G'$ gives a morphism $B_E G \to B_E G'$ (restriction of the action). 1:34:54

Definition. The sum $\coprod X_i$ of spaces is the disjoint union with each X_i open; a map out of it is continuous iff its restrictions are, which is the universal property of the coproduct. 1:36:59

Tape 2, side A

40 Sums and quotients of topoi

Proposition. *The product category* $\prod_i E_i$ *of topoi is a topos, and* $\mathrm{Hom}_{\mathrm{top}}(\prod_i E_i, F) \simeq \prod_i \mathrm{Hom}_{\mathrm{top}}(E_i, F)$*: it is the sum (coproduct) of the* E_i *in the 2-category of topoi, the projections being the inverse images of the inclusions* $E_i \to \prod E_i$*.* 1:40:43

Remark. Other limits and colimits of topoi (amalgamated sums, quotients) can be formed too; applied to spaces, a quotient taken in topoi may be well behaved where the quotient of spaces is wild (stated without proof). 1:40:43

41 Induced topoi

Definition. For an object X' of a topos E, the induced topos E/X' is the category of objects of E over X'; it plays the part of the space X' over the final object X (for $E = \mathrm{Top}(X)$ and X' étale over X, it is the topos of the space X'). 1:43:57

Proposition. *The restriction* $j^* \colon E \to E/X'$*,* $Y \mapsto Y \times X'$ *(viewed over* X' *by the second projection), commutes with colimits and finite limits; it is the inverse image of a morphism of topoi* $j \colon E/X' \to E$*, the localization morphism.* 1:45:57

Proposition. j^* *has a left adjoint* $j_!$*, which forgets the structure map to* X'*, as well as the right adjoint* j_**; the localization morphism is the analogue of an étale map of spaces, and of an open immersion when* X' *is a subobject of the final object.* 1:47:01

42 Sober spaces and generic points

Session of 4 June 1973

Video 2/4 (PqCQM3Pcl6Q): tape 1, side A (1:55:37–2:44:38); tape 1, side B (2:44:38–3:33:35); tape 2 (3:33:35–about 3:41:00).

The first half hour is a question session, as agreed at the end of the previous meeting. The lecture closes at 3:40:24 with the next meeting fixed for Wednesday.

Tape 1, side A

Definition. A space is sober if every irreducible closed subset has exactly one generic point. 1:55:37

Discussion. A question asks for a criterion characterising sober spaces, in the way sheaves of groups or rings might; the lecturer says he knows no such characterisation. 1:55:37

Proposition. *Spaces with the same sobrification have equivalent categories of sheaves. The sober space attached to a topos has the isomorphism classes of points as underlying set, with the topology given by the subobjects of the final object; for a space X it is the set of irreducible closed subsets of X with the evident topology.* 1:56:40

Example. For a finitely generated algebra A over an algebraically closed field K, the set of maximal ideals with the Zariski topology (the spaces of Serre's FAC) is not sober: irreducible closed subsets of positive dimension have no generic point. 1:57:43

Example. Hausdorff spaces and schemes are sober. 1:59:48

Example. For a non-separated topological group G, the closure of a point is a translate of the closure of the unit; the associated sober space is the separated quotient of G by $\overline{\{e\}}$, so passing to it identifies points but adds none. 2:00:48

Definition. On a set of ideals of a ring (maximal, prime, primary, …), the Zariski-type topology has as basic closed sets, for each subset S of the ring, the ideals of the given type containing S; the closed sets are the arbitrary intersections of finite unions of these. 2:02:48

Remark. Such spaces are generally non-separated, and specialization ($x \in \overline{\{y\}}$) is then an order which need not be an equivalence relation. 2:03:51

43 Tensor product of sheaves of modules

Definition. For sheaves of modules M, N, the presheaf $U \mapsto M(U) \otimes N(U)$ is in general not a sheaf; the tensor product $M \otimes N$ is its associated sheaf, which solves the universal problem for bilinear pairings. 2:04:56

Proposition. *If N is constant free of finite rank n, the presheaf tensor product is already a sheaf, equal to M^n; the same holds when N is a direct factor of such a module, $M \otimes N$ being the image of the induced projector of M^n.* 2:09:06

44 Algebras of finite presentation and passage to the limit

Proposition. *If A is a reduced algebra of finite type over $\mathbb{Z}$, its normalization is of finite type over $\mathbb{Z}$.* 2:17:22

Proposition. *If $K = \varinjlim K_i$ (filtering, K_i of finite type over $\mathbb{Z}$) and A is a K-algebra of finite presentation, then $A = A_i \otimes_{K_i} K$ for some i and some K_i-algebra A_i of finite presentation.* 2:19:28

Proposition. *For most geometric properties of* $\mathrm{Spec} A \to \mathrm{Spec} K$ *(geometrically connected, irreducible or normal fibres, non-empty fibres, surjective, flat, smooth, étale, unramified), A has the property iff A_j has it over K_j for all large j (EGA IV, chapter on limits).* 2:20:31

Proposition. *If A is of finite presentation and flat over K, the descent datum can be chosen with A_i flat over K_i.* 2:25:34

45 Points of presheaf topoi

Proposition. *For $c \in C$, evaluation $F \mapsto F(c)$ is a fibre functor on $\widehat{C}$, and $C \to \{$points of $\widehat{C}\}$ is fully faithful.* 2:30:00

Proposition. *Filtering inverse limits of points of $\widehat{C}$ exist, so $C \to \{$points$\}$ extends to an equivalence* $\mathrm{Pro}(C) \simeq \{$*points of* $\widehat{C}\}$. 2:32:06

Proposition. *A point of $\widehat{C}$ is essential (its fibre functor commutes with all limits) iff it is a retract of a representable; if C is idempotent complete, the essential points are exactly those coming from C.* 2:33:09

Proposition. *If $\widehat{C} \simeq \mathrm{Top}(X)$ for a space X, then C is an ordered category: there is at most one arrow between two objects.* 2:34:11

Definition. A category is ordered if it has at most one arrow between any two objects; it is then equivalent to the category of a preordered set, $x \leq y$ iff there is an arrow $x \to y$. 2:36:21

Proposition. *Conversely, if C is an ordered category, $\widehat{C}$ is equivalent to the topos of a space with underlying set* $\mathrm{Ob}\, C$. 2:38:25

Definition. The open sets of that space are the sieves $O \subset \mathrm{Ob}\, C$ ($x \in O$, $y \leq x \Rightarrow y \in O$), which correspond to the subobjects of the final object of $\widehat{C}$. 2:40:32

Proposition. *In this topology, the smallest open set containing x is $\{y : y \leq x\}$.* 2:43:40

Tape 1, side B

46 Topoi of ordered categories and of spaces

Proposition. *For an ordered category C with the sieve topology on* $\mathrm{Ob}\, C$*, the sets O_x form a basis of opens, $O_x \subset O_y$ iff $x \leq y$, so presheaves on the basis are the presheaves on C.* 2:44:38

Theorem. *For an ordered category C and its associated space, every presheaf on the basis is a sheaf (a basic open covered by basic opens is equal to one of them), so* $\mathrm{Top}(X_C) \simeq \widehat{C}$. 2:46:46

Proposition. *A point x of a sober space X is essential (its fibre functor commutes with arbitrary limits) iff x has a smallest open neighbourhood; if* $\mathrm{Top}(X) \simeq \widehat{C}$*, the essential points must form a conservative family.* 2:48:55

Definition. A subset $X' \subset X$ is very dense if every non-empty locally closed subset of X meets X'; this is equivalent to the fibre functors at the points of X' being conservative (two opens with the same trace on X' are equal). 2:50:59

Theorem. *For a sober space X, $\mathrm{Top}(X) \simeq \widehat{C}$ for some category C iff the set of essential points of X is very dense; C can then be taken to be that set ordered by specialization.* 2:53:08

47 Localization of topoi

Definition. For S' in a topos E, the morphism $E/S' \to E$ with inverse image $X \mapsto X \times S'$ is the localization morphism, and that inverse image the localization functor; a property of objects is said to hold locally if it holds after this base change. 2:57:17

Example. For $E = \mathrm{Top}(S)$ and S' an étale space over S, the localization morphism is the one induced by the map of spaces $S' \to S$. 2:57:17

48 Classifying topoi of subgroups as localizations

Proposition. *For a homomorphism $f : G \to G'$ of group objects of a topos E, the inverse image of $Bf : BG \to BG'$ restricts the action along f and commutes with all limits and colimits (computed in E).* 2:58:21

Theorem. *For a subgroup H of a group object G of a topos E, $BH \simeq BG/(G/H)$, and under this equivalence the morphism $BH \to BG$ is the localization morphism at the object G/H of BG (with G acting by left translation).* 3:01:29

Proposition. *The equivalence sends an object $A \to G/H$ of $BG/(G/H)$ to its fibre over the unit section, with the induced H-action, and an object F of BH to the contracted product $G \times^H F$ over G/H; these functors are quasi-inverse.* 3:04:38

Remark. The morphism $BH \to BG$ behaves like a covering rather than an embedding: the localization object $G/H \to 1$ is not a monomorphism. 3:22:52

49 Galois theory of coverings and classifying topoi

Theorem. *For a connected, locally connected and locally simply connected space Z with base point z, taking the fibre at z with its monodromy action is an equivalence from coverings of Z to $\pi_1(Z, z)$-sets; the inverse sends a G-set E to $P \times^G E$, P the universal covering based at z.* 3:23:52

Proposition. *Thus the classifying topos BG of a discrete group is the topos of coverings of a space Z with $\pi_1(Z, z) = G$.* 3:29:12

Proposition. *Subgroups $H \subset G$ correspond to pointed transitive G-sets G/H, hence to connected coverings $Z' \to Z$ with a point over z; and $BG/(G/H)$ is then the category of coverings of Z'.* 3:30:18

Tape 2

50 Subgroups as coverings

Proposition. *For the connected covering $Z' \to Z$ corresponding to $H \subset G$, the map $\pi_1(Z', z') \to \pi_1(Z, z) = G$ is the inclusion of H; so, up to homotopy, $BH \to BG$ is a covering* 3:33:35

with fibre G/H.

Theorem. *For a group G acting on a space Z and a subgroup $H \subset G$, $\mathrm{Top}(Z,H) \simeq \mathrm{Top}(Z,G)/(Z \times G/H)$ (G acting diagonally), the equivalence sending an object over $Z \times G/H$ to its part over $Z \times \{e\}$ with the induced H-action; $\mathrm{Top}(Z,H) \to \mathrm{Top}(Z,G)$ is the localization morphism.* 3:36:57

51 Adjoints $f_!$ and $f^!$ in examples

Session of 6 June 1973

Video 2/4 (PqCQM3Pcl6Q): tape 2 (from about 3:43:44); tape 1 (4:22:54–5:12:09).

Dating: the catalogue starts this meeting at 4:22:54, but the text runs on there in mid-sentence; the meeting opens, on the evidence of the text, with the question session at 3:43:44, after the closing of the previous one.

Tape 2 (from 3:43:44)

Proposition. *For a closed embedding $i\colon Y \to X$, i_* is fully faithful on abelian sheaves (an equivalence onto the sheaves vanishing off Y), and commutes with all limits and colimits.* 3:49:03

Proposition. *For a closed embedding i, on abelian sheaves i_* has a right adjoint $i^!$: $i^!F$ is the subsheaf of F of sections supported in Y, i.e. $\Gamma(U, i^!F) = \Gamma_{Y\cap U}(U,F)$.* 3:50:06

Remark. The right derived functors of $i^!$ are the sheaves of local cohomology. 3:52:18

Proposition. *For an open embedding $j\colon Y \to X$, j^* commutes with products, so $j_!$ exists on sheaves of sets: it views an étale space over Y as an étale space over X, with empty stalks off Y.* 3:54:19

Proposition. *For sheaves of abelian groups on an open $Y \subset X$, $j_!F$ is the extension by zero: $(j_!F)(U) \subset F(U \cap Y)$ consists of the sections whose support, closed in $U \cap Y$, is closed in U.* 3:56:27

Counterexample. The $j_!$ for abelian sheaves is not the $j_!$ for sheaves of sets: the underlying sheaf of sets of the extension by zero has non-empty stalks off Y, whereas the set-theoretic $j_!$ has empty ones. 4:00:38

Proposition. *For a continuous map f of spaces, f_* preserves the initial (empty) sheaf iff the image of f is dense; for topoi, iff the morphism is dominating, i.e. factors through no proper closed subtopos.* 4:03:45

Question. For a morphism of topoi, can the right adjoint $f^!$ of f_* (on sets, not on abelian objects) exist unless f is an equivalence? 4:05:50

Remark. On abelian sheaves $f^!$ is important (it underlies duality), and taking adjoints of the functors attached to a morphism of topoi often leads to further morphisms of topoi. 4:07:52

52 Local properties and the notion of site

Definition. A fibre space $E \to X$ is locally trivial if X has an open covering U_i with $E|_{U_i} \simeq U_i \times F_i$; being locally trivial is a property that is local on X, whereas being trivial is not. 4:10:06

Definition. For a group scheme G over K, X is a torsor under G if, for every K', $X(K')$ is empty or a principal homogeneous space under $G(K')$. 4:14:21

Remark. Such an X is trivial over K' iff $X_{K'}$ has a section (for a field K', iff it is non-empty); it is called locally trivial for a chosen class of admissible base changes (arbitrary or Galois extensions) if it becomes trivial after one of them. 4:16:27

Definition. A site is a category equipped with a notion of localization, here called a topology by abuse of language (not a topology on the objects and arrows). 4:18:37

Definition. A topology on a category C can be given, in a first form, by a set $\mathrm{Cov}(X)$ of covering families of morphisms $X_i \to X$ for each object X. 4:20:40

Tape 1

53 Coverings, pretopologies and local properties

Example. On algebras over a ring A, the families (B_i) for which some finite subfamily gives a faithfully flat map $A \to \prod B_i$ are the coverings of a (simplified) fpqc-type topology. 4:22:54

Definition. When C has fibre products, covering families are required to be stable under base change: if $(X_i \to X)$ covers X and $X' \to X$ is any morphism, then $(X_i \times_X X' \to X')$ covers X'. 4:25:05

Definition. Covering families are required to be transitive: if $(X_i \to X)$ covers X and each X_i is covered by $(X_{ij} \to X_i)_j$, then $(X_{ij} \to X)_{i,j}$ covers X. 4:25:05

Definition. A family consisting of a single isomorphism is covering; and covering families are saturated under refinement: if (X_i) covers X and each $X_i \to X$ factors over X through a member of a family $(X'_{i'})$, then $(X'_{i'})$ covers X. 4:27:07

Definition. A class of families satisfying base change, transitivity and the trivial axiom is a pretopology; saturating it under refinement gives a topology. 4:31:25

Remark. Giraud later replaced covering families by covering sieves, which makes the formulas simpler and removes the assumption that fibre products exist. 4:31:25

Definition. Given a category F_X attached contravariantly to each object X (inverse images), a property of a situation over X holds locally if it holds on the pulled-back situations over X_i for some covering $(X_i \to X)$; for instance, an arrow is locally a monomorphism (resp. an isomorphism). 4:34:36

Proposition. *For a topology generated by a pretopology, a property is local iff it holds after pullback along a covering family of the pretopology.* 4:36:48

54 Sheaves on a site

Definition. For a site C with fibre products, a presheaf F is separated if $F(X) \to \prod_i F(X_i)$ is injective for every covering, and a sheaf if $F(X) \to \prod_i F(X_i) \rightrightarrows \prod_{i,j} F(X_i \times_X X_j)$ is exact; $\widetilde{C} \subset \widehat{C}$ is the full subcategory of sheaves. 4:38:54

Theorem. *For a small site C, $\widetilde{C}$ is a topos (stated with technical provisos omitted).* 4:42:04

Remark. Many different sites give the same topos, as a group has many presentations: the category C plays the part of generators and the topology the part of relations. 4:42:04

Proposition. *$\widetilde{C}$ is the topos universal for functors out of $\widehat{C}$ commuting with colimits and finite limits that make covering families epimorphic; covering families are in general not epimorphic in $\widehat{C}$.* 4:45:14

55 The associated sheaf on a site

Theorem. *The inclusion $i\colon \widetilde{C} \to \widehat{C}$ has a left exact left adjoint a, the associated sheaf functor.* 4:55:43

Definition. For a presheaf F, $LF(X)$ is the filtering colimit, over the coverings $(X_i \to X)$, of the matching families $\ker\left(\prod F(X_i) \rightrightarrows \prod F(X_i \times_X X_j)\right)$; in terms of covering sieves $R \subset X$, $LF(X) = \varinjlim_R \mathrm{Hom}(R, F)$. 4:57:48

Proposition. *The functor $L\colon \widehat{C} \to \widehat{C}$ is left exact (a filtering colimit of finite limits), LF is separated for every F, and $\mathrm{Hom}(F,G) \simeq \mathrm{Hom}(LF,G)$ for every sheaf G.* 5:02:07

Proposition. *$F \to LF$ is a monomorphism iff F is separated, and an isomorphism iff F is a sheaf; if F is separated, LF is a sheaf. Hence $a = L \circ L$.* 5:04:18

Proposition. *$\widetilde{C}$ is closed in $\widehat{C}$ under limits; a colimit of sheaves is the associated sheaf of the colimit taken in $\widehat{C}$; exactness properties of $\widetilde{C}$ involving colimits and finite limits follow from those of $\widehat{C}$ since a is exact.* 5:07:34

Definition. The representable presheaf $\mathrm{Hom}(-,X)$ is a sheaf iff a morphism $Y \to X$ amounts to a compatible family of morphisms $Y_i \to X$ on a covering of Y; when this holds for all X (a subcanonical topology, a name not used in the lecture), $C \to \widehat{C}$ factors through $\widetilde{C}$. 5:09:50

56 Algebraic structures and exactness properties

Session of 20 June 1973

Video 2/4 (`PqCQM3Pcl6Q`): tape 1, side A (5:12:09–6:01:04); tape 1, side B (6:01:04–6:50:09).

Tape 1, side A

Remark. Algebraic structures (groups, rings, modules, comodules, group actions, torsors) can be expressed by direct and inverse limits and arrows between them, so they make sense in any category with the appropriate limits, subject to exactness conditions. 5:14:17

Example. A basic exactness property: filtering direct limits commute with finite inverse limits. 5:16:11

Remark. Topoi are, apart from infinite inverse limits, the categories with exactly the exactness properties of **Set**. 5:17:07

Question. Does every type of algebraic structure have a classifying topos containing a universal model, so that types of structure correspond to topoi, added axioms to subtopoi, and the homotopy type of the classifying topos gives a classifying space of the structure? (Programme stated as expected.) 5:18:59

Definition. For sets D' and D'' of diagrams, Cat_D is the 2-category of categories having limits of type D' and colimits of type D'', with functors commuting with them, and natural transformations as 2-morphisms. 5:21:59

Definition. An exactness property of type (I, Σ), with $I \in \mathrm{Cat}_D$ and Σ a set of arrows of I: a category $C \in \mathrm{Cat}_D$ satisfies it if every morphism $\varphi\colon I \to C$ of Cat_D sends the arrows of Σ to isomorphisms. 5:24:53

Definition. A 2-sum of objects I_λ of a 2-category is an object I with morphisms $I_\lambda \to I$ such that $\mathrm{Hom}(I, C) \to \prod_\lambda \mathrm{Hom}(I_\lambda, C)$ is an equivalence of categories for every C; it is determined up to equivalence, itself unique up to unique isomorphism. 5:33:53

Proposition. *If 2-sums exist in Cat_D, a conjunction of exactness properties $(I_\lambda, \Sigma_\lambda)$ is one exactness property (I, Σ), with I the 2-sum and Σ the union of the images of the Σ_λ, for the categories C admitting at least one D-functor $I_\lambda \to C$ for each λ (otherwise (I, Σ) may hold vacuously).* 5:35:46

Definition. The D-envelope of a category J is an object $\mathrm{Env}(J)$ of Cat_D with a functor $J \to \mathrm{Env}(J)$ such that, for every $C \in \mathrm{Cat}_D$, composition $\mathrm{Hom}_D(\mathrm{Env}(J), C) \to \mathrm{Hom}(J, C)$ is an equivalence of categories (a 2-universal problem). 5:41:42

Remark. Asserted without proof: if D' or D'' is small, every category has a D-envelope; it is the category obtained by adjoining the limits of type D' and the colimits of type D''. 5:43:33

Discussion. A remark suggests 2-representability as a weaker substitute for representability when no genuine reflection exists; the reply is inconclusive. 5:45:39

Counterexample. The naive fibre product of categories (pairs of objects, resp. arrows, with equal images in S) is not invariant under equivalence: replacing X and Y by equivalent categories mapping into two isomorphic copies of S makes it empty. 5:46:41

Definition. The 2-fibre product $X \times^2_S Y$ of $f\colon X \to S$ and $g\colon Y \to S$ has as objects the triples (x, y, α) with $\alpha\colon f(x) \xrightarrow{\sim} g(y)$, and as arrows the pairs of arrows compatible with the α's. 5:51:25

Proposition. *If $X \to S$ is fibred, the naive fibre product is equivalent to the 2-fibre product.* 5:51:25

Proposition. *The 2-fibre product is 2-universal:* $\mathrm{Hom}(C, X \times^2_S Y) \to \mathrm{Hom}(C, X) \times^2_{\mathrm{Hom}(C,S)} \mathrm{Hom}(C, Y)$ *is an equivalence of categories, for every category C.* 5:54:21

Definition. In any 2-category, a 2-fibre product of X, Y over S is an object P with morphisms to X and Y and an isomorphism of the composites to S, such that this functor on Hom-categories is an equivalence for every object; it is determined up to equivalence, unique up to unique isomorphism. 5:57:10

Definition. 2-limits of categories are taken of pseudo-functors $I \to$ **Cat**, for which composition holds only up to coherent isomorphism. 5:59:09

Tape 1, side B

57 Envelopes, localization and a worked exactness property

Remark. All the universal problems considered are 2-universal, their solutions defined up to equivalence; 2-inverse limits are taken of pseudo-functors $I \to$ (a 2-category), and 2-direct limits by reversing arrows. 6:01:04

Discussion. Asked whether the one-dimensional construction of coproducts would also be valid in the 2-categorical setting, the lecturer does not know and says the construction is ugly even at the lower level. 6:03:07

Remark. Size conventions: a universe U is fixed; categories are required to have U-small Hom-sets, but may be large, such as the categories of all U-groups or U-rings. 6:05:59

Example. Universality of sums as an exactness property: if X is the sum of objects x_α and $X' \to X$ is any morphism, then with $X'_\alpha = x_\alpha \times_X X'$, the object X' is the sum of the X'_α. 6:11:00

Proposition. *This property is expressed by inverting arrows: realisations of the diagram type in C are the functors from its envelope $\tilde{J}$ to C; those for which the squares are Cartesian and X is the sum are the functors inverting a set Σ of arrows of $\tilde{J}$; and the conclusion says that the canonical arrow $\coprod X'_\alpha \to X'$ is also inverted.* 6:17:58

Definition. For $J \in \mathrm{Cat}_D$ and a set Σ of arrows of J, $J[\Sigma^{-1}] \in \mathrm{Cat}_D$ is defined by the 2-universal property that $\mathrm{Hom}_D(J[\Sigma^{-1}], C) \to \mathrm{Hom}_D(J, C; \Sigma)$ is an equivalence onto the functors commuting with D-limits that invert Σ. 6:20:49

Remark. Existence of $\tilde{J}$ and $J[\Sigma^{-1}]$ is assumed; he expects it in all cases except possibly when neither D' nor D'' is small. 6:24:36

Definition. An elementary exactness property is given by a diagram Δ, diagrams Δ'_i (inverse limit type) and Δ''_j (direct limit type) with cones to vertices x'_i, x''_j of Δ, and further diagrams Λ'_α, Λ''_β with vertices y'_α, y''_β: whenever a realisation of Δ in C makes each x'_i the limit and each x''_j the colimit of the corresponding diagram, then each y'_α is a limit and each y''_β a colimit of the corresponding diagram. 6:25:34

Discussion. A doubt is raised about exactness properties holding vacuously because no suitable functors exist; the lecturer first dismisses it, then concedes the point, since constant diagrams need not satisfy the imposed conditions. 6:31:21

Proposition. *An elementary exactness property becomes a functor condition: with Σ the arrows of $\tilde{\Delta}$ comparing the vertices with the limits and colimits, realisations are the functors $\tilde{\Delta}[\Sigma^{-1}] \to C$, and one requires that they invert the further comparison arrows for y'_α and y''_β.* 6:35:10

Definition. For a set η of exactness properties, $\mathrm{Cat}_{D,\eta} \subset \mathrm{Cat}_D$ is the full sub-2-category of categories satisfying them, with the same functors. 6:38:51

Definition. An algebraic structure T realisable in the categories C of a given class assigns to each C the category $T(C)$ of realisations of T in C, functorially in the functors $C \to C'$ commuting with the relevant limits. 6:44:44

Example. The structure of group makes sense in any category with finite products; a product-preserving functor sends group objects to group objects. 6:46:39

Proposition. *An object P with an action of a group G is a torsor iff $G \times P \to P \times P$ is an isomorphism and $P \to 1$ is an epimorphism; so a functor preserves torsors only if it preserves epimorphisms.* 6:47:38

Proposition. *A morphism $P \to Q$ is a monomorphism iff the square $P = P \times_Q P$ is Cartesian, and an epimorphism iff the dual square is co-Cartesian; so preserving epimorphisms is an exactness property (commuting with certain pushouts).* 6:48:34

58 Limits and colimits in the 2-category of D-categories

Session of 27 June 1973

Video 2/4 (PqCQM3Pcl6Q): tape 2, side A (6:50:09–7:38:43); tape 2, side B (7:38:43–8:27:21); tape 1, side A (8:27:21–9:02:23).

Dating: the catalogue puts the first two sides under 20 June, but the meeting opens at 6:52:32 with a recap of the previous week and the announcement that Wednesday 4 July is a holiday, which places it in the week of 27 June; the last side, labelled 27 June, continues it.

Tape 2, side A

Definition. Standing assumptions on D: (1) for each category I, the functor $C \mapsto \mathrm{Hom}(I, C)$ on Cat_D is 2-representable by an object $\tilde{I}$; (2) for $C \in \mathrm{Cat}_D$ and a set Σ of arrows of C, $R \mapsto \mathrm{Hom}_D(C, R; \Sigma)$ is 2-represented by $C[\Sigma^{-1}] \in \mathrm{Cat}_D$, which is not the plain localization in **Cat**. 6:57:23

Proposition. *Under these assumptions* Cat_D *has all 2-inverse limits, computed as in* **Cat***: limits of type D', D'' and the exactness properties are checked componentwise.* 7:02:13

Definition. The 2-inverse limit of a pseudo-functor $I \to$ **Cat** is the category of Cartesian sections of the associated fibred category over I. 7:06:10

Proposition. Cat_D *has all 2-direct limits: take the 2-direct limit L in* **Cat** *(the fibred category F with Cartesian arrows inverted), pass to $\tilde{L}$, then invert the set Σ of arrows comparing the images of the D-limits of the C_i with the limits in $\tilde{L}$.* 7:11:55

Example. Finite inverse limits with left exact functors (structures defined by finite inverse limits). 7:19:51

Example. Arbitrary inverse limits with limit-preserving functors, related to the embedding $C \to \widehat{C}$ and to viewing a structure on an object as a structure on the functor it represents. 7:20:47

Example. Preordered categories (at most one arrow between two objects) with finite lower bounds and arbitrary upper bounds; the functors are the monotone maps preserving them. 7:23:39

Example. Semi-additive categories (initial and final objects agree, sums agree with products) with additive functors; additive categories with an extra condition. 7:25:29

Example. Abelian categories with exact functors; the universal abelian category generated by one object is $\tilde{I}$ for I the one-point category. 7:26:25

Example. Abelian categories in which filtering direct limits exist and are exact, with functors preserving them (for instance Ind-categories). 7:27:23

Example. Categories with finite inverse and direct limits, with functors preserving them; with the exactness properties of topoi, the pretopoi (corresponding to coherent topoi). 7:28:28

Example. Topoi: D' the finite diagrams, D'' all small diagrams (so D'' is not a small set), with the existence of generators imposed separately; the corresponding 2-category of topoi is too large. 7:29:23

Question. Are the essential topoi, i.e. topoi in which arbitrary products of epimorphisms are epimorphisms and arbitrary products commute with sums, exactly the presheaf topoi $\widehat{C}$ (when they have generators)? 7:31:23

Example. Ordered sets with finite lower bounds and arbitrary upper bounds, with monotone maps preserving them; and those in which finite lower bounds distribute over arbitrary upper bounds (distributivity of products over sums). 7:34:14

Definition. An algebraic theory T for a fixed D is a pseudo-functor $C \mapsto T(C)$ from Cat_D to **Cat**, $T(C)$ being the category of realisations of T in C, with functors $\mathrm{Hom}_D(C, C') \to \mathrm{Hom}(T(C), T(C'))$ compatible with composition. 7:36:14

Tape 2, side B

59 Representable theories and constructions

Definition. A theory T is representable by $R \in \mathrm{Cat}_D$ if $T(C) \simeq \mathrm{Hom}_D(R, C)$, naturally in C; R then carries a universal realisation of T, every realisation in C being its image under an essentially unique D-functor $R \to C$. 7:38:43

Definition. Theories over D (pseudo-functors $\mathrm{Cat}_D \to$ **Cat**), their morphisms (families of functors $T(C) \to T'(C)$ compatible with the transition functors) and the transformations between these form a 2-category, with hom-categories $\underline{\mathrm{Hom}}(T, T')$. 7:41:39

Proposition. *(Yoneda) If T is represented by R, then $\underline{\mathrm{Hom}}(T, T') \simeq T'(R)$ for every theory T'.* 7:43:33

Definition. A construction on T is a morphism of theories $T \to \tau$, where $\tau(C) = C$ is the identity theory: a way of producing, functorially in C, an object of C from a realisation of T in C (for groups: the centre, the abelianization, the n-th power). 7:44:34

Proposition. *If T is represented by R, the category $\underline{\mathrm{Hom}}(T, \tau)$ of constructions on T is equivalent to R.* 7:44:34

Proposition. *The category of constructions on any theory lies in Cat_D: its limits of type D' and D'', and its exactness properties, are computed objectwise.* 7:49:24

Proposition. *Writing R_T for the category of constructions on T, there is a canonical functor $T(C) \to \mathrm{Hom}_D(R_T, C)$, natural in C; thus T_{R_T} plays the part of a representable envelope of T.* 7:54:27

60 Base objects and the rules for algebraic theories

Definition. A theory T has base objects indexed by a small set I if there is a faithful morphism of theories $T \to \tau^I$, where $\tau^I(C) = C^I$; the realisations are then families of objects of C with extra structure. 7:59:16

Proposition. *If T is represented by R, morphisms $T \to \tau^I$ correspond to families $(R_i)_{i \in I}$ of objects of R (equivalently to D-functors $\tilde{I} \to R$), and faithfulness means that $\mathrm{Hom}_D(R, C) \to C^I$ is faithful for all C; one then says that I D-generates R.* 8:03:22

Remark. Expected, not proved: if I generates R, the objects of R are obtained from the base objects by a bounded number of limit operations on the arrows already constructed. 8:08:14

Definition. Rule 1 (empty structure): for a small set I, $T(C) = C^I$ is an algebraic theory with I as base objects, represented by $\tilde{I}$. 8:11:04

Definition. Rule 2 (adding axioms): if T is algebraic and represented by R, and Σ is a set of arrows of R (arrows between constructions), the theory $T'(C) \subset T(C)$ of realisations whose D-functor $R \to C$ inverts Σ is algebraic, and is represented by $R[\Sigma^{-1}]$. 8:13:57

Example. For groups, commutativity is an axiom of this kind: the canonical arrow from G to its abelianization is required to be an isomorphism. 8:15:54

Proposition. *When the needed diagrams are in D' or D'', the following are axioms of this kind: an arrow between constructions is a monomorphism (its diagonal is an isomorphism), is an epimorphism, or two parallel arrows between constructions are equal (equalizer or coequalizer an isomorphism).* 8:16:52

Counterexample. "X acts faithfully on Y" cannot be expressed by inverting arrows, even in a topos: it is not stable under base change. 8:18:46

Counterexample. "An object of a topos is locally constant with n sheets" cannot be so expressed; the corresponding theories are not algebraic, not representable. 8:19:42

Definition. Rule 3 (adding arrows): if T is algebraic and represented by R, and H is a small set with source and target maps $\alpha_1, \alpha_2 \colon H \to \mathrm{Ob}\, R$ (formal arrows between constructions), $T'(C)$ is the category of pairs (ψ, β) with $\psi \in T(C)$ and, for each $h \in H$, an arrow $\beta(h) \colon \psi(\alpha_1 h) \to \psi(\alpha_2 h)$ in C; $T'(C) \to T(C)$ is faithful, no longer full. 8:20:37

Tape 1, side A

61 Theories as 2-fibred categories over Cat_D

Question. Are the algebraic theories relative to D, to be defined by stepwise rules, exactly the theories 2-represented by objects of Cat_D? For D the 2-category of topoi, this would identify types of algebraic structure with classifying topoi. 8:34:57

Question. Can every exactness property be written as "every D-functor $I \to C$ factors through a given D-functor $I \to I'$", i.e. $\mathrm{Hom}_D(I', C) \to \mathrm{Hom}_D(I, C)$ fully faithful with prescribed image? Only the converse direction (an exactness property gives such a pair, with $I' = I[\Sigma^{-1}]$) was shown. 8:38:47

Discussion. The audience asks whether every condition that a functor between categories be an equivalence can be rephrased as an exactness property; the lecturer proved only one direction and doubts the converse without extra hypotheses. 8:41:40

Example. "Every monoid object is a group" is an exactness property: a monoid is a group iff the shear map $(x, y) \mapsto (x, xy)$ of $G \times G$ is an isomorphism (Cartesian squares in the nerve). 8:44:36

Discussion. Examples are tried for exactness properties: that every monoid in a category is a group is expressible through Cartesian squares, whereas that every semigroup is a monoid seems not to be. 8:44:36

Question. Is "every semigroup object is a monoid" an exactness property? 8:46:26

Remark. A theory is identified with the family of its realisations $C \mapsto T(C)$, a 2-fibred category over Cat_D, not with a presentation by axioms: different axiomatisations of the same structure give the same theory. 8:47:38

Remark. Constructions that are functorial only for isomorphisms, not for homomorphisms, do not fit the framework. 8:55:09

Counterexample. In a topos, "X over B is locally constant of type I_n" (locally isomorphic to the constant object with n elements) makes sense intrinsically but defines no algebraic theory: it needs the object $\underline{\mathrm{Isom}}(I_{n,B}, X)$, not expressible by limits alone. 8:57:03

Counterexample. For rings, "the spectrum is connected" is expressible by these means, but "the spectrum has exactly n connected components" is not, being tied to local constancy. 8:58:56

62 Adding arrows and combining theories

Session of late June 1973 (catalogue: 27 June)

Video 2/4 (`PqCQM3Pcl6Q`): tape 2, side A (9:02:23–9:52:24); tape 2, side B (9:52:24–10:14:40).

Dating: this meeting begins with no recap, carrying out what the previous one announced for its next session, and it ends by moving the following class from Tuesday to Monday; the tape gives no day, so the catalogue date of 27 June is kept as a label only.

Tape 2, side A

Example. Extra structure on a group G by adding arrows: a map $G^{\mathrm{ab}} \to G \times G$, or $G^3 \to G^4$, constructed from G in any category of Cat_D. 9:02:23

Proposition. *If T is represented by R, the theory obtained by adding the arrows H is represented by $R_1[\Sigma^{-1}]$, where R_1 is the envelope of the diagram type $R \cup H$ with the relations from composition imposed, and Σ the arrows expressing the failure of exactness.* 9:04:24

Definition. Rule 4 (combining theories): for a small family of algebraic theories T_α with base objects I_α, the product $\prod_\alpha T_\alpha$ is algebraic, with base objects $\coprod I_\alpha$ (a group and a ring together); it is represented by the 2-sum of the R_α in Cat_D (disjoint union, D-envelope, then inversion of the arrows expressing exactness). 9:12:18

Proposition. *The 2-inverse limit of a pseudo-functor $\alpha \mapsto T_\alpha$ of algebraic theories (computed objectwise on $T_\alpha(C)$) is represented by the 2-direct limit of the R_α in Cat_D.* 9:17:10

Example. The 2-fibre product of the theory of groups and the theory of rings over the theory of Lie rings (via the Lie ring of the descending central series, resp. the commutator bracket) is the theory of a group and a ring with an isomorphism between their Lie rings. 9:19:11

Proposition. *For theories $T_\alpha \to \tau^I$ over the same base objects, the 2-fibre product over τ^I (all the structures on the same objects) is represented by the 2-amalgamated sum of the R_α over $\tilde{I}$; for instance, a group and a ring on the same underlying object.* 9:21:06

Definition. A theory is algebraic if it is obtained from the empty structure by iterating rules 2, 3 and 4, transfinitely if necessary. 9:23:03

Proposition. *Every algebraic theory is representable, by an R having a small D-generating family (constructions by generators and relations).* 9:24:58

Proposition. *If R is small, T_R is algebraic: take $I = \mathrm{Ob}\, R$, start from τ^I, add the arrows of R as formal arrows, impose that composites and identities are respected (arrows becoming equal) and that the functor commutes with D-limits (arrows becoming isomorphisms).* 9:26:59

Proposition. *Making given pairs of arrows equal is itself a solvable 2-universal problem in Cat_D (impose the equalities in* **Cat***, pass to the D-envelope, invert the comparison arrows); so rule 2 admits two kinds of axioms: arrows becoming isomorphisms and arrows becoming equal.* 9:32:49

Question. If R is not small and I is a small family of objects with $\mathrm{Hom}_D(R, C) \to C^I$ faithful for all C, is T_R algebraic? 9:39:31

Tape 2, side B

63 Generating families: limits of the criterion

Proposition. *If R is a lattice and D', D'' are the finite meets and joins, $\mathrm{Hom}_D(R, C)$ is a preordered category, so every functor out of it is faithful and the faithfulness condition on I is empty.* 9:52:24

Counterexample. For R the ordered category of ordinals below the first ordinal outside the universe, the empty family satisfies the faithfulness condition, yet the theory is not algebraic 9:55:22

(asserted, expected rather than proved); faithfulness of $\mathrm{Hom}_D(R, C) \to C^I$ is too weak a notion of generation.

Question. Is it enough to require that $I \to R$ factors through no proper full D-subcategory $R' \subset R$ (so that R is reached by transfinitely adding limits and inverting arrows from I)? 9:57:20

Question. Which conditions on a pair (R, I) make T_R an algebraic theory with base objects I, built from the empty structure on I by adding arrows and axioms? 10:00:19

Question. For which D are the envelope and localization problems solvable? Known for finite inverse limits with left exact functors, and for topoi; open for additive, abelian and Karoubian categories (where every projector has an image). 10:03:06

64 Additive categories as an exactness property

Discussion. The lecturer asks for examples of enveloping categories, additive or abelian; the room points to a universal additive category, but he stresses that his version also requires initial and final objects to coincide. 10:04:02

Definition. A semi-additive category has finite products and sums, the canonical arrow from the initial to the final object is an isomorphism (so zero morphisms exist), and the canonical arrow $X \sqcup Y \to X \times Y$ is an isomorphism. 10:05:52

Proposition. *In a semi-additive category every object carries a unique monoid structure and a unique comonoid structure, both commutative (Eckmann–Hilton), and the Hom-sets are commutative monoids.* 10:07:45

Proposition. *A semi-additive category is additive iff every monoid object is a group, which is an exactness property: the map $(x, y) \mapsto (x, x + y)$ of $X \times X$ is an isomorphism.* 10:09:37

Remark. No simple construction of an enveloping additive category is known when both direct and inverse limits occur; for finite inverse limits only, functors into abelian groups give one. 10:12:40

65 Algebraic theories and strict cogeneration

Session of 2 July 1973

Video 3/4 (4gynb9uTEE4): tape 1 (0:00:00–0:48:52); tape 2 (0:48:52–1:54:02); tape 3, side A (1:54:02–2:43:03); tape 3, side B (2:43:03–3:31:57).

Tape 1

Proposition. *For Δ = categories with arbitrary limits, the localization $C \to C[\Sigma^{-1}]$ taken in Δ is bijective on objects: adding axioms adds arrows, not constructions.* 0:04:18

Proposition. *A category I is strictly cogenerating in its envelope $\widetilde{I}$ under arbitrary limits.* 0:05:21

Definition. $J \subset E$ is "strictly generating" if every object of E is a colimit of a diagram of objects of J; example: J in $\widehat{J}$. 0:06:26

Proposition. *If I is strictly cogenerating in each R_α of a filtered system in Δ, it is so in its 2-colimit taken in Δ.* 0:07:31

Proposition. *Strict cogeneration is claimed to be transitive (false, retracted on 10 July, 3:39:45); it is preserved by adjoining arrows in the sense of Δ and by localization in Δ.* 0:10:48

Proposition. *At each step of the construction of an algebraic theory with base I, the image of I remains strictly cogenerating (retracted on 10 July, 3:37:36: false already for groups).* 0:11:52

Theorem. *For $\Delta =$ categories with arbitrary limits, the algebraic theories are those represented by an $R \in \Delta$ having a small strictly cogenerating subcategory (retracted on 10 July, 3:43:00: the condition is necessary, not sufficient).* 0:12:56

Proposition. *Moreover, $I \to R$ is basic if and only if the image of I is strictly cogenerating (retracted on 10 July, 3:37:36); dually, R^{op} has arbitrary colimits and a small strictly generating set.* 0:14:08

Proposition. *If $I' \subset R$ is a small full strictly cogenerating subcategory, the Δ-functor $\widetilde{I'} \to R$ is essentially surjective and full; R is a category of fractions of $\widetilde{I'}$.* 0:18:24

Proposition. *Such a theory is obtained in three steps (empty structure on I, adjoining arrows, axioms), without transfinite construction (retracted on 10 July, 3:41:54: a transfinite construction may be needed).* 0:22:37

Proposition. *If T is represented by R having a small strictly cogenerating subcategory, $T(\mathbf{Set}) = \mathrm{Hom}_{\Delta}(R, \mathbf{Set}) \simeq R^{\mathrm{op}}$: T is determined by its realizations in sets.* 0:30:45

Discussion. The lecturer notes that universal problems for other exactness conditions can have natural solutions too large to lie in the universe, while for toposes the solutions are explicit and well behaved. 0:35:03

Example. For the theory of groups, $R \simeq \mathbf{Grp}^{\mathrm{op}}$; the universal object is $\mathbb{Z}$, a cogroup of **Grp** via $\mathbb{Z} \to \mathbb{Z} * \mathbb{Z}$, $x \mapsto x_1 x_2$. 0:37:13

Definition. For C small and a class D' of diagrams, the envelope of C under limits of type D' is the smallest full subcategory of the envelope $\widetilde{C}$ containing C and stable under these limits. 0:44:41

Example. On categories with finite products, the theory of groups is represented by a small category, the full subcategory of $\mathbf{Grp}^{\mathrm{op}}$ of finitely generated free groups. 0:45:46

Counterexample. A theory on categories with limits of type D' defines by restriction one on categories with arbitrary limits; there is no inverse operation in general. 0:46:53

66 Accessible functors

Tape 2

Proposition. *If $F : E \to E'$ is π-accessible and sends E_π into E'_π (π-accessible objects), F is the extension Ind_π of its restriction $E_\pi \to E'_\pi$.* 0:51:15

Proposition. *For F accessible and π large enough, $F(E_\pi) \subset E'_{\pi^{\pi_0}}$ (π_0 fixed, SGA 4); since $(2^\lambda)^{\pi_0} = 2^\lambda$ for $\lambda \geq \pi_0$ infinite, $F(E_\pi) \subset E'_\pi$ if $\pi = 2^\lambda$.* 0:52:15

Proposition. *An accessible functor between accessible categories respects the canonical filtration by π-accessible objects for arbitrarily large π.* 0:54:23

Proposition. *For $i : C \hookrightarrow E$ small full and E' with colimits, the restriction $i^* : \mathrm{Hom}(E, E') \to \mathrm{Hom}(C, E')$ has a fully faithful left adjoint $i_!$.* 0:56:38

Proposition. *For E accessible, $F : E \to E'$ is accessible if and only if it is in the essential image of $i_!$ for some small $C \subset E$.* 0:57:40

Proposition. *For C small with split idempotents, $\mathrm{Ind}_\pi(C)$ has arbitrary colimits if and only if C has colimits of cardinality at most π.* 0:59:46

Proposition. *For such a C, $\mathrm{Ind}_\pi(C)$ is the Δ_π–Δ envelope of C:* $\mathrm{Hom}_\Delta(\mathrm{Ind}_\pi C, F) \simeq \mathrm{Hom}_{\Delta_\pi}(C, F)$ *for every F with arbitrary colimits.* 1:03:11

67 Representability and accessible categories

Proposition. *The categories R representing algebraic theories are exactly the "co-accessible" categories: $S = T(\mathbf{Set}) \simeq R^{\mathrm{op}}$ is accessible (and cocomplete).* 1:06:26

Proposition. *On a cocomplete accessible category E, a functor $E^{\mathrm{op}} \to \mathbf{Set}$ is representable if and only if it sends colimits to limits.* 1:09:46

Proposition. *A functor $F : E \to \mathbf{Set}$ (E cocomplete accessible) is representable if and only if it commutes with arbitrary limits and is accessible.* 1:10:49

Proposition. *SGA 4: a functor $F : E \to \mathbf{Set}$ is pro-representable by a pro-object with small indexing if and only if it is left exact and accessible.* 1:12:06

Proposition. *If E has arbitrary limits, a pro-representable functor $E \to \mathbf{Set}$ is representable if and only if it commutes with these limits.* 1:13:11

Proposition. *Between cocomplete accessible categories, F has a right adjoint if and only if it commutes with colimits, and a left adjoint if and only if it commutes with limits and is accessible.* 1:14:14

Example (SGA 4). On **Grp**, a functor with small values, pro-represented by a non-small system of simple groups, commutes with limits without being accessible. 1:15:31

68 Morphisms of algebraic theories

Discussion. An example from SGA of a functor commuting with inverse limits yet not accessible, hence not representable: built from an inverse system of arbitrarily large simple groups, with small values on each group. 1:15:31

Proposition. *A morphism of algebraic theories $T \to T'$ is equivalent to a functor $S' \to S$ commuting with colimits; its right adjoint $S \to S'$ is $T(\mathbf{Set}) \to T'(\mathbf{Set})$.* 1:17:44

Proposition. *The functors $S \to S'$ arising from morphisms of algebraic theories are exactly those that commute with limits and are accessible.* 1:20:53

Proposition. *If $F \to I$ is fibred over I small, with accessible fibres and transition functors, the category of cartesian sections (2-limit) and that of all sections are accessible.* 1:22:11

Proposition. *For E accessible and I small, $\mathrm{Hom}(I, E)$ is accessible, as is $\mathrm{Hom}(I[I^{-1}], E)$, the functors sending every arrow to an isomorphism.* 1:24:21

Proposition. *The 2-colimit of a small system of cocomplete accessible categories, taken in cocomplete categories and cocontinuous functors, is accessible.* 1:28:40

Question. Is the 2-colimit in **Cat** of a small system of accessible categories accessible? 1:31:03

Example. The coproduct in **Cat** of two nonempty categories with colimits has no initial object, nor a coproduct of an object of each. 1:32:09

Proposition. *For a morphism of algebraic theories, $T(\mathbf{Set}) \to T'(\mathbf{Set})$ has a left adjoint (free structure); example: the enveloping group of a monoid.* 1:45:06

Proposition. *If a fully faithful functor admits a left adjoint, this adjoint is a localization (passage to a category of fractions).* 1:47:14

Proposition. *The full subtheories of $T_{R'}$ correspond to the localizations $R' \to R'[\Sigma'^{-1}]$ in Δ, Σ' stable under limits, and to the reflective full subcategories of S' stable under limits.* 1:48:19

Proposition. *For an algebraic subtheory, Σ' is generated, by limit operations, by a small set of arrows.* 1:50:27

Question. Is a full subcategory S of an accessible category S', stable under limits and whose inclusion admits a left adjoint, accessible? 1:51:31

69 Constructing algebraic theories

Tape 3, side A

Proposition. *Limit-passage clause: the 2-limit, over T^I, of a filtered system of algebraic theories with base I is algebraic with base I.* 1:57:30

Proposition. *Representable theories form a full 2-subcategory of the Δ-theories, 2-equivalent to Δ^{op} via $R \mapsto T_R$; it contains the algebraic theories.* 1:58:34

Definition. R is algebraic with (discrete) base I if it is the end of a transfinite sequence in Δ starting from the envelope of I, each step inverting, equalizing or adding arrows, with 2-colimits at limit ordinals. 2:00:40

Counterexample. Faithfulness of $\mathrm{Hom}_\Delta(R, C) \to C^I$ for every C does not suffice for $I \to R$ to be basic (known counterexample). 2:03:54

Question. "Problem three" (open): is $I \to R$ basic if and only if every strictly full subcategory of R stable under limits of type D and containing the image of I is equal to R? 2:03:54

Proposition. *This condition is equivalent to: the transfinite saturation of the image of I under limits of type D' and D'' reaches R.* 2:06:04

Proposition. *If D' and D'' are essentially small (and if problems one and two preserve smallness), the algebraic objects of Δ are exactly the small ones.* 2:08:14

Question. Conjectural statement: if R is algebraic, one can enlarge I so that no arrows remain to be added; T_R is then a full subtheory of $C \mapsto \mathrm{Hom}(I', C)$. 2:10:23

70 Representability of theories

Proposition. *Necessary condition for 2-representability: T commutes with 2-limits, $T(\varprojlim C_i) \simeq \varprojlim T(C_i)$.* 2:17:56

Proposition. *On a category with arbitrary limits, a functor to* **Set** *that commutes with these limits is pro-representable by a projective system, possibly non-small.* 2:18:59

Proposition. *If T is 2-representable, T sends faithful (resp. fully faithful) Δ-functors to faithful (resp. fully faithful) functors.* 2:20:10

71 Exactness properties of theories

Definition. Exactness property given by $R \in \Delta_D$ and Σ: every Δ_D-functor $R \to C$ inverts Σ; equivalent to: $\mathrm{Hom}_{\Delta_D}(R', C) \to \mathrm{Hom}_{\Delta_D}(R, C)$ is an equivalence, R' the localization. 2:23:27

Definition. "Generalized exactness properties": for a given Δ_D-functor $R \to R'$, require that $\mathrm{Hom}(R', C) \to \mathrm{Hom}(R, C)$ be an equivalence, fully faithful or faithful. 2:25:38

Proposition. *The 2-subcategories of Δ_D so defined are stable under 2-limits in* **Cat**. 2:26:42

Example ($D = \varnothing$). Requiring the arrow of $\{0 \to 1\}$ to be inverted defines groupoids. 2:28:51

Example. Equalizing the two arrows of $\{0 \rightrightarrows 1\}$ defines preordered categories, equivalent to preordered sets, 2-equivalent to ordered sets. 2:29:58

Example. The two conditions together define essentially discrete categories, 2-equivalent to **Set**. 2:32:06

Example. For $\Delta =$ **Cat**: the envelope of I is I; problem two is solved by the category of fractions $I[\Sigma^{-1}]$ (Gabriel–Zisman). 2:33:10

Example. For groupoids: the envelope of I is its fundamental groupoid $I[I^{-1}]$, with the same connected components; problem two is trivial. 2:34:16

Categories of fractions are in Gabriel and Zisman's book. [7] 2:34:16

Example. For preorders: the envelope of I is $\mathrm{Ob}\, I$ with the preorder $x \geq y$ if there is an arrow $x \to y$. 2:36:21

Example. Making relations $x_i \leq y_i$ of a preorder invertible: take the preorder generated by the given preorder and the $y_i \leq x_i$. 2:37:24

Example. For discrete categories: the envelope of I is $\pi_0(I)$, the set of connected components; problem two is trivial. 2:38:30

Proposition. *If $T = T_R$ and if C has limits of type δ commuting with colimits of type D'', $T(C)$ has these limits, computed argumentwise.* 2:41:54

Tape 3, side B

Proposition. *A morphism of representable theories $T \to T'$ gives $T(C) \to T'(C)$ commuting with these limits; likewise $T(C) \to T(C')$ if $C \to C'$ preserves them.* 2:45:15

Proposition. *For every object r of R, the evaluation functor $r_C : T(C) \to C$ commutes with limits of type δ.* 2:46:18

Example. The category of torsors under a group $G \neq 1$ has no products, no coproducts, and no initial or terminal object. 2:47:20

Proposition. *If $D'' = \varnothing$ and if C has limits of type δ (resp. arbitrary), $T(C)$ has them too.* 2:48:23

Proposition. *For $\Delta' \subset \Delta$ (fewer limits required), restriction sends Δ-theories to Δ'-theories and algebraic theories to algebraic theories.* 2:50:40

Proposition. *If problems one and two have a solution in Δ', the restriction of T_R to Δ' is represented by the "Δ–Δ' envelope" R' of R.* 2:52:50

Proposition. *R' is obtained from the envelope of R in Δ' by inverting the comparison arrows of the limits of type D of R.* 2:56:04

Counterexample. The restriction functor $\mathrm{Hom}(T_1, T_2) \to \mathrm{Hom}(T_1', T_2')$ is in general neither faithful, nor full, nor essentially surjective. 3:01:55

72 Envelopes and categories of fractions

Proposition. *For J small, $J \to \widehat{J}$ is the envelope under colimits: for D cocomplete, restriction of cocontinuous functors $\widehat{J} \to D$ to J is an equivalence.* 3:10:48

Proposition. *Every presheaf F on J is the colimit over J/F of the objects of J over F; the extension is $\widehat{\varphi}(F) = \varinjlim_{J/F} \varphi(X)$.* 3:11:49

Proposition. *Problem one for Δ = categories with small limits: the envelope of a small I is $\mathrm{Hom}(I, \mathbf{Set})^{\mathrm{op}}$, with the Yoneda embedding.* 3:14:59

Proposition. *In Δ, equalizing $f, g : X \rightrightarrows Y$ amounts to inverting the kernel $\mathrm{Ker}(f, g) \to X$.* 3:17:04

Proposition. *Problem two: if Σ' is the saturation of Σ, the category of fractions $C[\Sigma'^{-1}]$ taken in* **Cat** *is in Δ, and $C \to C[\Sigma'^{-1}]$ is the solution.* 3:18:06

Proposition. *The saturation Σ' (arrows inverted by every Δ-functor inverting Σ) contains the isomorphisms, is stable under composition and base change, and if two of u, v, vu are in it, so is the third.* 3:21:33

Proposition. *If $u : X \to Y$ is in Σ', the diagonal $X \to X \times_Y X$ is in Σ'.* 3:23:39

Proposition. *Σ' admits a calculus of fractions: $\mathrm{Hom}(X, Y) = \varinjlim \mathrm{Hom}_C(X', Y)$, over the filtered projective system of the $X' \to X$ in Σ'.* 3:24:44

Proposition. *If C has finite limits, so does $C[\Sigma'^{-1}]$, and the localization functor is left exact.* 3:27:53

Proposition. *Σ' is stable under arbitrary fibre products: if the $X_i \to Y$ are in Σ', so is $\prod_Y X_i \to Y$.* 3:28:59

Proposition. *$C[\Sigma'^{-1}]$ has arbitrary products, images of those of C; thus $C \to C[\Sigma'^{-1}]$ is a Δ-functor.* 3:31:07

73 Further remarks on algebraic theories

Session of 10 July 1973

Video 3/4 (4gynb9uTEE4): tape 1, side A (3:31:57–4:36:29); tape 1, side B (4:36:29–5:41:33).

Tape 1, side A

Counterexample. For E in Δ and a non-small saturated class $\widetilde{\Sigma}$, the category of fractions $E[\widetilde{\Sigma}^{-1}]$ may have non-small Homs, and is then not in Δ. 3:35:28

Counterexample. For an algebraic theory R with base objects I, I is not in general strictly cogenerating in R: a group structure is not determined by the maps $x \mapsto x^n$. 3:37:36

Counterexample. Strict generation is not transitive: $\mathbb{Z}$ is strictly generating in free groups, these are so in **Gr**, but $\{\mathbb{Z}\}$ is not in **Gr** (likewise in **Ab**). 3:39:45

Counterexample. An algebraic theory relative to limits is not always obtained in two steps (arrows, then axioms); a transfinite construction, indexed by a small ordinal, may be needed. 3:41:54

Discussion. The lecturer retracts a theorem stated earlier: a small strictly cogenerating family is necessary for an algebraic theory but not sufficient, and he previews a counterexample involving the opposite of the category of sets. 3:41:54

Proposition. *If $R \in \Delta$ is algebraic, R admits a small strictly cogenerating family; the converse is false.* 3:43:00

Example. $\mathbf{Set}^{\mathrm{op}}$ admits a strict generator in the sense of SGA 4 (the two-element set: every set is an equalizer of two maps between powers of 2), though not in the sense of 0:06:26, but is the category $T(\mathbf{Set})$ of no algebraic theory T. 3:44:08

Accessible functors: he refers again to SGA 4. [3] 3:45:13

Proposition. $\mathbf{Set}^{\mathrm{op}}$ *is equivalent to the category of complete atomic Boolean algebras, via* $X \mapsto \mathcal{P}(X)$. 3:46:17

Discussion. A listener objects that the opposite of sets is the category of complete atomic Boolean algebras; the lecturer replies that its axioms need a large family, so it is not algebraic here. 3:46:17

Remark. Completeness of a Boolean algebra concerns families indexed by an arbitrary set I: a non-small family of operations and axioms, excluded from the algebraic theories of the course. 3:52:47

Discussion. Debate on whether joins over every index set count as a small set of axioms: the lecturer insists on smallness, others suggest bounding cardinalities, and he calls the choice partly one of taste. 3:59:22

Proposition. *Every complete Boolean algebra falls, for a suitable cardinal* π*, under the theory of* π*-complete Boolean algebras, which has only a small set of operations.* 4:07:05

Discussion. The lecturer admits long-standing unease with definitions quantifying over arbitrary sets, notes that no contradiction has been shown, and sets aside non-small infinitary operations as a different framework. 4:08:07

Proposition. *The objects of* Δ *representing algebraic theories are the* Δ*-envelopes (adjoining small limits) of small categories of* Δ_π *(limits of cardinality* $\leq \pi$*).* 4:10:16

Proposition. *The categories* $S = R^{\mathrm{op}} = T(\mathbf{Set})$ *of set-theoretic realizations of these theories are accessible categories.* 4:11:21

74 Accessible categories

Definition. A preordered set I "dominates π" if it is filtered and every family of cardinality $\leq \pi$ of elements of I has an upper bound; for $\pi \geq 2$, the second condition implies the first. 4:14:44

Definition. E satisfies L_π if colimits indexed by every small I dominating π exist in E; E satisfies L if it satisfies L_π for some small cardinal π. 4:15:52

Definition. $F\colon E \to E'$ is π-accessible if E satisfies L_π and F commutes with colimits indexed by the I dominating π; accessible if it is π-accessible for some small π. 4:18:02

Definition. An object X of E is π-accessible (resp. accessible) if the functor $\mathrm{Hom}(X, -)$ from E to **Set** is π-accessible (resp. accessible). 4:21:13

Definition. E is π-accessible if it has a small full strictly generating subcategory C consisting of π-accessible objects (every object is a colimit of a diagram in C). 4:23:29

Remark. This definition is stronger than that of SGA 4, I, 9, which only required a small generating subcategory. 4:24:37

Accessible functors: SGA 4 once more. [3] 4:24:37

Proposition. *If* E*,* F *satisfy* L_π*, the full subcategory* $\mathrm{Hom}_\pi(E, F)$ *of* π*-accessible functors is stable in* $\mathrm{Hom}(E, F)$ *under colimits computed argumentwise.* 4:26:48

Proposition. *If $F = \mathbf{Set}$ (or, more generally, if in F colimits along an I dominating π commute with such limits), it is also stable under limits, computed argumentwise, indexed by a small category J with at most π arrows.* 4:27:56

Proposition. *In* **Set**, *colimits along an I dominating π commute with limits along a category with at most π arrows.* 4:30:13

Proposition. *Accessible functors form a full subcategory of* $\mathrm{Hom}(E, F)$ *stable under small colimits and (for $F = \mathbf{Set}$) small limits computed argumentwise.* 4:30:13

Proposition. *If E satisfies L_π, the full subcategory E_π of π-accessible objects is stable under colimits indexed by a category with at most π arrows.* 4:31:16

Proposition. *If E satisfies L, the full subcategory of accessible objects of E is stable under small colimits.* 4:32:21

Proposition. *If E is accessible, every object of E is accessible.* 4:33:26

75 Examples of accessible categories

Tape 1, side B

Definition. A small subcategory C is generating by universal strict epimorphisms if every object is the target of a universally strictly epimorphic family from C; this is stronger than strictly generating. 4:36:32

Proposition. *In a topos, every epimorphic family is strictly epimorphic and universal.* 4:37:34

Proposition. *If E satisfies L, has fibre products, kernels $(u, v) \mapsto \mathrm{Ker}(u, v)$ forming an accessible functor, and such a C, then every object of E is accessible and E is accessible (SGA 4, I, 9.11).* 4:37:34

Example. Every topos is an accessible category. 4:39:42

Counterexample. A set X of cardinality ≥ 2 is not an accessible object of $\mathbf{Set}^{\mathrm{op}}$; hence $\mathbf{Set}^{\mathrm{op}}$, which has small colimits and a generator, is not accessible. 4:39:42

Discussion. The lecturer argues that the opposite of the category of sets is not co-accessible in the relevant sense, since large products break the commutation that is required. 4:40:45

Proposition. *For π infinite, the subsets of cardinality $\leq \pi$ of a set I form an ordered set dominating π, and $\prod_{i \in I} Y_i$ is the limit of the corresponding partial products.* 4:41:48

Proposition. *If* $\mathrm{card}\, I > \pi$ *and* $\mathrm{card}\, Y_i \geq 2$, *the indicator of a point of $\prod Y_i$ factors through no partial product; the opposite of a nondegenerate topos, or of* **Ab**, *is not accessible.* 4:45:02

Question. If E and E^{op} are accessible, is E equivalent to the point category? 4:48:17

76 Ind-categories

Definition. For C small, $\mathrm{Ind}(C)$ is the category of filtered inductive systems in C, identified via $\varinjlim$ with a full subcategory of $\widehat{C}$. 4:50:29

Proposition. $\mathrm{Hom}_{\mathrm{Ind}(C)}((X_i), (Y_j)) = \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j)$ *(SGA 4, I, 8).* 4:52:37

Accessible functors: SGA 4 once more. [3] 4:53:40

Proposition. *$C \to \mathrm{Ind}(C)$ is universal: for E with filtered colimits, restriction to C of the functors* $\mathrm{Ind}(C) \to E$ *commuting with these colimits is an equivalence with* $\mathrm{Hom}(C, E)$. 4:54:44

Proposition. *If C has finite colimits, $\mathrm{Ind}(C)$ has small colimits and, for E having them, $F \mapsto F|_C$ identifies the functors commuting with these colimits with the right exact functors $C \to E$.* 4:58:00

Definition. $\mathrm{Ind}_\pi(C)$ is the full subcategory of $\mathrm{Ind}(C)$ of the systems indexed by an I dominating π; for C small, it is π-accessible. 5:00:08

Proposition. *An object of $\mathrm{Ind}_\pi(C)$ is π-accessible if and only if it is a direct summand of an object of C; a direct summand of a π-accessible object is again so.* 5:05:31

Definition. C is Karoubian if every idempotent has an image summand; $\mathrm{Ind}(C)$ and $\mathrm{Ind}_\pi(C)$ do not change if C is replaced by its Karoubian envelope. 5:06:36

Proposition. *If C is Karoubian and $E = \mathrm{Ind}_\pi(C)$, then $C \simeq E_\pi$ and $E \simeq \mathrm{Ind}_\pi(E_\pi)$.* 5:07:40

77 Cardinal filtrations

Proposition. *For $\pi' \geq \pi$, $\mathrm{Fil}^{\pi'} = \mathrm{Ind}_{\pi,\pi'}(C)$ (systems indexed by an I dominating π with $\mathrm{card}\, I \leq \pi'$) filters $\mathrm{Ind}_\pi(C)$ by essentially small subcategories, and $\mathrm{Fil}^\pi \simeq C$.* 5:08:44

Proposition. *The objects of $\mathrm{Fil}^{\pi'}$ are π'-accessible; if $\pi'^\pi = \pi'$, every π'-accessible object of $\mathrm{Ind}_\pi(C)$ is a direct summand of an object of $\mathrm{Fil}^{\pi'}$.* 5:09:49

Definition. "Cardinal filtration": a family $(\mathrm{Fil}^{\pi'})_{\pi' \geq \pi}$ of essentially small subcategories, stable under $\varinjlim$ of index I dominating π, $\mathrm{card}\, I \leq \pi'$, every object of $\mathrm{Fil}^{\pi''}$ being a colimit of objects of $\mathrm{Fil}^{\pi'}$ with index dominating π', of cardinality $\leq \pi''$. 5:10:52

Proposition. *The $\mathrm{Ind}_{\pi,\pi'}(C)$ form a cardinal filtration of $\mathrm{Ind}_\pi(C)$ if one restricts to cardinals such that $\pi''^{\pi'} = \pi''$ (bound $\pi''^{\pi'}$ of SGA 4, I, 9).* 5:14:07

Theorem. *If E has small colimits, the following are equivalent: a) E is π-accessible; b) $E \simeq \mathrm{Ind}_\pi(C)$, C small; c) E_π is essentially small and strictly generating.* 5:15:11

Proposition. *If $C \subset E_\pi$ is small and full, the extension $\mathrm{Ind}_\pi(C) \to E$ is fully faithful; it is an equivalence if C is strictly generating and stable under colimits with at most π arrows.* 5:22:51

Proposition. *A colimit indexed by a small category I is the colimit, along the ordered set (dominating π) of subcategories of I with at most π arrows, of the partial colimits.* 5:23:53

Proposition. *If E is π-accessible and has small colimits, $E \simeq \mathrm{Ind}_{\pi'}(E_{\pi'})$ for every $\pi' \geq \pi$.* 5:28:22

Counterexample. A π-accessible functor does not necessarily send π-accessible objects to π-accessible objects (example: E a point). 5:31:52

Proposition. *If E, E' are π_0-accessible with small colimits, $F : E \to E'$ is accessible if and only if there exist $\pi \geq \pi_0$ and $f : E_\pi \to E'_\pi$ with $F \simeq \mathrm{Ind}_\pi(f)$.* 5:32:58

78 Strictly generating functors

Session of 11 July 1973

Video 3/4 (4gynb9uTEE4): tape 1 (5:41:33–6:46:44); tape 2, side A (6:46:44–7:37:01).

Tape 1

Proposition. *For $i\colon C \to E$, $\varinjlim_{(Y,u)\in C/X} i(Y) \to X$ is an isomorphism for every X if and only if $E \to \widehat{C}$, $X \mapsto \mathrm{Hom}(i(-), X)$, is fully faithful: i is then called strictly generating.* 5:43:46

Proposition. *Then every X is the target of a strictly epimorphic family $i(Y_\alpha) \to X$; the converse holds if i is fully faithful and E has fibre products and universal strictly epimorphic families.* 5:46:59

Proposition. *If E has small colimits and C is small, i extends to $i_!\colon \widehat{C} \to E$, left adjoint to $E \to \widehat{C}$; i is strictly generating if and only if $i_!$ makes E a category of fractions of $\widehat{C}$.* 5:50:08

Theorem. *A category with small colimits has a small strictly generating subcategory if and only if it is equivalent to a reflective full subcategory of some $\widehat{C}$, C small.* 5:52:19

Example. $\mathbf{Set}^{\mathrm{op}}$ (complete atomic Boolean algebras) and $\mathbf{Ab}^{\mathrm{op}}$ (compact abelian groups) have small colimits; $\mathbf{Set}^{\mathrm{op}}$ has a small strictly generating family if and only if there is no proper class of measurable cardinals (Isbell), which ZFC does not decide. 5:53:21

Discussion. Whether compact spaces admit a small strictly generating family: a listener proposes ultrafilter operations, but the lecturer argues the point cannot strictly generate, since the forgetful functor to sets is not fully faithful. 5:54:23

Counterexample. In the category of compact spaces, the one-point space is not strictly generating: the underlying-set functor is faithful but not fully faithful. 5:57:44

79 Theories defined by finite limits

Discussion. The lecturer warns that a statement of the previous day was false: exactness conditions on objects and on functors must match, since existence conditions without uniqueness give no existence theorems for envelopes. 6:02:10

Proposition. *For Δ_D (limits of type D' only), the envelope of C is the subcategory of $\mathrm{Hom}(C, \mathbf{Set})^{\mathrm{op}}$ generated by C under iterated limits of type D'; C embeds fully in it.* 6:05:23

Proposition. *If R_0 represents a theory for Δ_D, the theory extended to Δ is represented by $R = T(\mathbf{Set})^{\mathrm{op}}$, the envelope of R_0 under small limits; $R_0 \subset R$ is full.* 6:07:34

Proposition. *A theory relative to Δ_D is equivalent to the pair (S, Σ_0), $S = T(\mathbf{Set})$, $\Sigma_0 = R_0^{\mathrm{op}} \subset S$, where S is the envelope of Σ_0 under small colimits.* 6:08:38

Proposition. *For D' consisting of the finite diagrams, the algebraic theories are the small categories R_0 with finite limits; dually, Σ_0 is small with finite colimits.* 6:10:44

80 Finitely presented objects

Definition. X is finitely presented ($\aleph_0$-accessible) if $\mathrm{Hom}(X, -)$ commutes with filtered colimits; we write S_{fp} for the full subcategory of these objects. 6:12:54

Proposition. *For a theory defined by finite limits, $\Sigma_0 \simeq S_{\mathrm{fp}}$ and $S \simeq \mathrm{Ind}(\Sigma_0)$; S and Σ_0 determine each other, and $R_0 \simeq (S_{\mathrm{fp}})^{\mathrm{op}}$.* 6:13:58

Proposition. *For $\Sigma_0 \subset S$ full, $\mathrm{Ind}(\Sigma_0) \to S$ is fully faithful if $\Sigma_0 \subset S_{\mathrm{fp}}$ (not conversely); if Σ_0 is stable under finite colimits, it is an equivalence if and only if Σ_0 is strictly generating.* 6:15:01

Example. In categories of groups or modules, the finitely presented objects are the groups or modules finitely presented in the usual sense. 6:16:10

Theorem. *S with small colimits is $T(\mathbf{Set})$ for a theory defined by finite limits if and only if S_{fp} is essentially small and strictly generating.* 6:17:18

Example. In **Gr**, for C full on the object $\mathbb{Z}$, $\widehat{C}$ consists of the sets acted on by the monoid $(\mathbb{Z}, \times)$; $G \mapsto (G, x \mapsto x^n)$ is faithful but not full: $\mathbb{Z}$ is generating, but not strictly. 6:19:26

Remark. For $C \subset E$ full, $E \to \widehat{C}$ fully faithful, faithful, conservative, or conservative on monomorphisms: distinct notions of generation, which coincide in a topos. 6:22:40

Discussion. Clarifying generating versus strictly generating, the lecturer lists several notions that coincide in a topos but differ in general; free groups of finite type serve as the natural strictly generating family for groups. 6:22:40

Proposition. *Finitely generated free groups form a small strictly generating subcategory of* **Gr**. 6:25:47

Proposition. *The morphisms $T \to T'$ of theories defined by finite limits correspond to the functors $S' \to S$ commuting with small colimits and preserving finitely presented objects.* 6:26:51

81 Lawvere theories and finite products

Proposition. *For finite products, the envelope of an object X is $(\mathbf{Set}_{\mathrm{f}})^{\mathrm{op}}$, with objects X^n; for products of cardinality $\leq \pi$, the opposite of the sets of cardinality $\leq \pi$.* 6:31:05

Definition. A theory with one base object defined by finite products is a category with objects $\mathbb{N}$, n being the n-th power of 1, given by the $H_n = \mathrm{Hom}(X^n, X)$ and their substitution (Lazard's "analyseurs"). 6:34:22

Definition. The realizations of such a (Lawvere) theory in a category E with finite products are the functors to E commuting with finite products. 6:36:31

Remark. For a set I of base objects, one starts from the opposite of the category of finite sets over I (objects $\prod_i X_i^{n_i}$, n_i almost all zero). 6:37:33

Counterexample. For a theory defined by finite products, Σ_0 is stable under finite coproducts and strictly generating in S, but this does not characterize the pairs (S, Σ_0). 6:38:37

Theorem. *S with small limits is the category of models of a theory with finite products and base objects I if and only if its equivalence relations are effective and I, consisting of finitely presented objects, detects effective epimorphisms (*$\mathrm{Hom}(i, f)$ *surjective).* 6:40:44

Proposition. *Then $S \to \mathbf{Set}^I$, $X \mapsto (\mathrm{Hom}(i, X))_{i \in I}$, has a left adjoint and is monadic ("tripleable").* 6:43:53

82 Birkhoff's theorem

Tape 2, side A

On algebraic theories he points to Lawvere, and to a paper of his own for tripleability. [8] 6:46:04

Theorem. *Birkhoff's theorem: a full subcategory of a category of structures defined by finite products, stable under products, subobjects and quotients, is defined by equations.* 6:50:08

Discussion. On Birkhoff's theorem, a subcategory closed under products, subobjects and quotients is defined by equations; torsion-free abelian groups inside abelian groups show that closure conditions alone do not capture all subcategories. 6:50:08

Proposition. *Such a subcategory is reflective: the adjoint associates to X its largest quotient belonging to the subcategory.* 6:51:11

Counterexample. Torsion-free abelian groups, stable under products and subgroups but not under quotients, are not definable by equations in **Ab**. 6:52:17

Proposition. *To a Birkhoff subcategory $S' \subset S$ corresponds $R \to R'$ bijective on objects and surjective on arrows: T' is deduced from T by equalizing arrows.* 6:54:26

Counterexample. Groups are the monoids in which a certain arrow is invertible; not stable under submonoids, they are not definable by equations in monoids. 7:05:11

Discussion. Groups inside monoids cannot be cut out by equations alone, since the integers contain the natural numbers as a submonoid that is not a group, so the subclass is not closed under submonoids. 7:05:11

Proposition. *The reflective full subcategories $S' \subset S$ such that the composite $S' \to \mathbf{Set}^I$ is monadic form a larger class than that of Birkhoff subcategories.* 7:07:29

83 Pseudo-topoi and classifying topoi

Definition. A "pseudo-topos" satisfies Giraud's axioms a), b), c): small colimits and finite limits; universal disjoint coproducts; universal effective equivalence relations. 7:10:46

Proposition. *The functors considered commute with small colimits and finite limits: they are the inverse images f^* of morphisms of topoi in the opposite direction.* 7:12:55

Definition. A topos $\mathcal{E}$ classifies a theory T, with universal object $U \in T(\mathcal{E})$, if $f \mapsto f^*U$ is an equivalence $\mathrm{Top}(\mathcal{F}, \mathcal{E})^{\mathrm{op}} \simeq T(\mathcal{F})$ for every $\mathcal{F}$ (2-arrows oriented by the direct images). 7:14:08

Proposition. *For C small with finite limits, the envelope of C among pseudo-topoi is $\widehat{C}$, with the Yoneda embedding.* 7:19:36

84 Continuous functors and morphisms of sites

Definition. $f : C \to C'$ between sites is continuous if $F \mapsto F \circ f$ sends sheaves on C' to sheaves on C; this depends only on $\varepsilon_{C'} \circ f : C \to \widetilde{C'}$. 7:21:48

Proposition. *By composition with ε_C, the functors $\widetilde{C} \to \widetilde{C'}$ commuting with small colimits are identified with the continuous functors $C \to \widetilde{C'}$.* 7:22:55

Proposition. *If C, C' have finite limits and f commutes with them, f is continuous if and only if it sends covering families to covering families.* 7:25:04

Example. For $X' \to X$ continuous, the inverse image of open sets is such a continuous functor from the site of open sets of X to that of X'. 7:25:04

Proposition. *In general, f is continuous if and only if, for every covering sieve R of X, $\varepsilon_{C'} f(X) = \varinjlim_{Y \in C/R} \varepsilon_{C'} f(Y)$.* 7:26:07

Proposition. *A sieve R of X is covering if and only if $\varepsilon(X) = \varinjlim_{Y \in C/R} \varepsilon(Y)$ in $\widetilde{C}$.* 7:27:12

Proposition. *If C, C' have finite limits and the topology of C' is coarser than the canonical one, a continuous f induces a morphism of topoi if and only if f is left exact.* 7:30:21

Definition. A morphism of sites is a continuous functor that induces a morphism of topoi; it is oriented in the direction of that morphism. 7:31:25

Proposition. *For a site C with finite limits and a topos $\mathcal{E}$, $\mathrm{Top}(\mathcal{E}, \widetilde{C})^{\mathrm{op}}$ is equivalent to the left exact functors $C \to \mathcal{E}$ sending covering families to epimorphic families.* 7:32:31

Proposition. *$\widehat{C}$ is $\widetilde{C}$ for the coarsest topology (covering families: those having a member with a section).* 7:34:39

Proposition. *If C has finite limits, $\mathrm{Top}(\mathcal{E}, \widehat{C})^{\mathrm{op}}$ is equivalent to the left exact functors $C \to \mathcal{E}$.* 7:34:39

85 Classifying topoi of finite-limit theories

Session of 12 July 1973

Video 4/4 (`ORG14LNb5gE`): tape 1 (0:00:00–0:49:57); tape 2, side A (0:49:57–1:38:22); tape 2, side B (1:38:22–2:26:53); tape 3, side A (2:26:53–3:15:55); tape 3, side B (3:15:55–4:05:06); tape 4, side A (4:05:06–4:53:49); tape 4, side B (4:53:49–5:36:07).

Tape 1

Theorem. *If C is a small category with finite limits, $\widehat{C}$ classifies the theory T_C: $\mathrm{Hom}_{\mathrm{Top}}(\mathcal{E}, \widehat{C}) \simeq \mathrm{Hom}_{\mathrm{lex}}(C, \mathcal{E})$ for every topos $\mathcal{E}$.* 0:00:05

Example. The classifying topos of groups is the category of covariant functors from finitely presented groups to **Set**. 0:02:11

Example. The classifying topos of commutative rings is that of functors from finitely presented rings to **Set**, i.e. of presheaves on affine schemes of finite type over $\mathbb{Z}$. 0:03:15

Proposition. *The envelope $\widetilde{C}$ of C under finite limits is the opposite of the category of finitely presented functors $C \to$ **Set**; $C \to \widetilde{C}$ is fully faithful.* 0:05:25

Proposition. *If $\mathcal{E}$ has finite limits, $\mathrm{Hom}(C, \mathcal{E}) \simeq \mathrm{Hom}_{\mathrm{lex}}(\widetilde{C}, \mathcal{E})$.* 0:06:28

Definition. Enveloping topos of a small category C: the topos of presheaves on $\widetilde{C}$; the morphisms from a topos $\mathcal{E}$ to it are equivalent to $\mathrm{Hom}(C, \mathcal{E})$. 0:07:34

86 Subtopoi and localization

Remark. Localization: for a set Σ of arrows of a topos $\mathcal{E}$, one seeks a universal Δ-functor $\mathcal{E} \to \mathcal{E}[\Sigma^{-1}]$ making the arrows of Σ invertible. 0:10:03

Proposition. *The subtopoi of $\mathcal{E}$ correspond bijectively to the saturated sets of arrows of $\mathcal{E}$ (stable under limits and local in nature) (SGA 4, exercise).* 0:12:16

Definition. A morphism of topoi $i\colon \mathcal{E}' \to \mathcal{E}$ is an embedding if i_* is fully faithful, i.e. if i^* is a localization functor. 0:13:21

Proposition. *The subtopoi of $\mathcal{E}$ are identified with the strictly full subcategories whose inclusion admits a left exact left adjoint.* 0:15:30

Proposition. *For a small category C, the topologies J on C correspond bijectively, via $J \mapsto \mathrm{Sh}(C, J)$, to the subtopoi of $\widehat{C}$.* 0:16:37

Proposition. *A continuous map $f\colon X \to Y$ induces an embedding of topoi iff the topology of X is the inverse image of that of Y; subspaces of Y give subtopoi, but not all of them.* 0:17:39

Proposition. *A morphism $\mathcal{F} \to \mathcal{E}$ factors through a subtopos $\mathcal{E}'$ iff its direct image factors through $\mathcal{E}'$, iff its inverse image factors through the localization $\mathcal{E} \to \mathcal{E}'$.* 0:18:44

Proposition. *To a subtopos $\mathcal{E}'$ one associates the arrows made invertible by i^*; to Σ, the objects X such that $\mathrm{Hom}(Z, X) \to \mathrm{Hom}(Y, X)$ is bijective for every $Y \to Z$ in Σ.* 0:19:49

Theorem. *$\mathcal{E}[\Sigma^{-1}]$ exists and is a subtopos of $\mathcal{E}$; every subtopos is obtained in this way by inverting a family of arrows.* 0:22:00

Proposition. *If $\mathcal{E} = \mathrm{Sh}(C, J)$, its subtopoi correspond to the topologies on C finer than J; likewise for the subtheories of the theory represented by $\mathcal{E}$.* 0:23:02

87 Generating subcategories of a topos

Proposition. *If a small full subcategory C generates the topos $\mathcal{E}$, $\mathcal{E}$ is a subtopos of $\widehat{C}$: $\mathcal{E} \simeq \mathrm{Sh}(C, J)$ for the induced topology.* 0:32:47

Proposition. *For C full in $\mathcal{E}$: (a) $\mathcal{E} \to \widehat{C}$ fully faithful is equivalent to (b) every X is a colimit of the Y in C/X; (b) implies (c): every X is the target of a strictly epimorphic family from C.* 0:37:28

Discussion. The lecturer corrects an SGA claim on strictly generating families: a family with sources in a subcategory need not make the subcategory strictly generating, as the free groups on one and two generators show. 0:37:28

Counterexample. Counterexample to (c) $\Rightarrow$ (b): in groups, with $C = \{\mathbb{Z}\}$, $\mathbb{Z} * \mathbb{Z}$ satisfies (c) but is not a colimit of $C/(\mathbb{Z} * \mathbb{Z})$. 0:39:40

Counterexample. Enlarging a strictly epimorphic family can make it lose this property (all the arrows $\mathbb{Z} \to \mathbb{Z} * \mathbb{Z}$ in groups). 0:40:46

Proposition. *If fibre products exist and distribute over coproducts, $(X_i \to X)$ is strictly epimorphic iff $\coprod X_i \to X$ is; this is false in general.* 0:42:52

Discussion. A correction on strictly epimorphic families: the equivalence with the epimorphism of their sum needs fibre products distributing over sums, and the repaired statement asks for universality as well. 0:42:52

Proposition. *Corrected statement: (c′) every X is the target of a family from C that is strictly and universally epimorphic; then (c′) implies (b).* 0:46:05

Proposition. *If a strictly and universally epimorphic family factors through a family $(X_i \to Y)$, the latter is again strictly and universally epimorphic.* 0:47:09

Proposition. *(b) is equivalent to (b′): for every X, the family of all arrows $Y \to X$, Y in C, is strictly epimorphic.* 0:48:11

Tape 2, side A

Proposition. *A family $(Y \to X)$ is strictly epimorphic iff for every Z, $\mathrm{Hom}(X, Z)$ is identified with the compatible families of elements of the $\mathrm{Hom}(Y, Z)$ (sheaf condition).* 0:50:01

88 Algebraic theories and topoi

Theorem. *$R \in \Delta$ represents an algebraic theory iff R is a topos; $B\colon I \to R$ is basic iff the finite products of the $B(i)$ generate R.* 1:04:03

Proposition. *Every topos R represents an algebraic theory: if $C \subset R$ is small, generating and stable under finite limits, T_R is a full subtheory of $T_C = \mathrm{Hom}_{\mathrm{lex}}(C, -)$.* 1:06:58

Proposition. *For such a C, the morphisms of topoi $\mathcal{E} \to R$ correspond to the left exact functors $C \to \mathcal{E}$ sending covering families to epimorphic families.* 1:10:36

Proposition. *If $R' \to R$ induces for every topos $\mathcal{E}$ a fully faithful embedding $T_{R'}(\mathcal{E}) \to T_R(\mathcal{E})$, R' is a subtopos of R (exercise of SGA 4).* 1:11:47

Proposition. *A subtopos of $R = \mathrm{Sh}(C, \tau)$ is $\mathrm{Sh}(C, \tau')$ with τ' finer; its theory is deduced from T_R by a small set of axioms.* 1:19:23

89 Limits of topoi

Proposition. *Arbitrary (not necessarily filtered) 2-limits of topoi exist; in Δ, they are 2-colimits.* 1:23:42

Remark. Construction: $C_i \subset R_i$ compatible small generating subcategories, C the 2-colimit of the C_i in **Cat**, then localization of the enveloping topos of C at a small Σ. 1:29:00

Proposition. *The 2-colimit of a system of small categories is obtained by inverting the cartesian arrows of the associated (co)fibred category.* 1:30:05

Proposition. *A 2-limit of accessible categories, with accessible transition functors, is accessible.* 1:36:31

90 Algebraic theories built from finite products

Tape 2, side B

Proposition. *The category J_0 freely generated by finite products over a set I is the opposite of the families of finite sets indexed by I, almost all empty; it is also the envelope under finite limits of the discrete I.* 1:38:26

Proposition. *$R_0 = \widehat{J_0}$ classifies families of objects indexed by I: $\mathrm{Hom}_\Delta(R_0, \mathcal{E}) \simeq \mathcal{E}^I$.* 1:40:37

Proposition. *Adding axioms: $R_{\alpha+1} = R_\alpha[\Sigma_\alpha^{-1}]$ has the same objects as R_α, so J_0 still generates it.* 1:41:40

Remark. Adding arrows given by a graph $D = (H \rightrightarrows S)$: $R_{\alpha+1}$ is the 2-pushout in Δ of R_α and $\rho(\widetilde{D})$ over $\rho(S)$ (ρ the enveloping topos, $\widetilde{D}$ the free category). 1:43:52

Proposition. *In this construction, every object of $R_{\alpha+1}$ is covered by images of objects of R_α; hence J_0 generates $R_{\alpha+1}$.* 1:51:23

Proposition. *Limit step: for α a limit, R_α is the 2-colimit in Δ of the R_β, $\beta < \alpha$; J_0 still generates it.* 1:56:57

Proposition. *Necessity: if $I \to R$ is basic for an algebraic theory, R is a topos generated by the finite products of objects of I.* 1:59:07

Proposition. *Sufficiency: if the finite products of the objects of I generate the topos R, the theory is obtained without transfinite recursion: finite products, arrows between them, then axioms.* 2:00:13

Proposition. *For J small with finite products, the theory $\mathcal{E} \mapsto \mathrm{Hom}_\Pi(J, \mathcal{E})$ of functors preserving finite products is represented by a topos $\rho_{\Pi,\Delta}(J)$.* 2:03:18

Proposition. *If the subcategory $J_2 \subset R$ of finite products of objects of I generates R, R is a subtopos of $\rho_{\Pi,\Delta}(J_2)$; two successive localizations make it one.* 2:08:52

Proposition. *Preservation of finite products is an axiom: the arrow of constructions $\varepsilon_{X\times Y} \to \varepsilon_X \times \varepsilon_Y$ must become an isomorphism.* 2:16:31

Proposition. *If $R \in \Delta$ represents a theory, its category of constructions is R; equalizing two arrows amounts to inverting a kernel, hence a single type of axiom.* 2:21:54

Tape 3, side A

Proposition. *In a topos, $u\colon X \to Y$ is an epimorphism iff $Y \sqcup_X Y \to Y$ is an isomorphism; "being covering" is thus written as an invertibility axiom.* 2:28:03

91 Classifying topos of an object and its subtopoi

Example. The classifying topos of an object without structure is the category of functors from finite sets to **Set** (presheaves on $J = \{X^n\}$). 2:33:54

Proposition. *If B_T classifies the theory T, its category of points is equivalent to T(**Set**).* 2:36:05

Proposition. *For all topoi $\mathcal{E}$ and B, $\mathrm{Hom}_\Delta(B, \mathcal{E})$ admits filtered colimits, since in a topos they commute with finite limits.* 2:37:13

Proposition. *The essential points of the classifying topos of objects correspond to finite sets.* 2:39:28

Proposition. *The smallest subtopos containing a family of points of a topos is defined by the arrows whose stalks at these points are isomorphisms.* 2:40:31

Example. The subobjects of the terminal object of the classifying topos B of objects are $\varnothing$, the image of $X \to e$, and $e\colon B$ has three open subtopoi. 2:42:38

Example. The closed complement Φ_0 of B^* classifies initial objects; it is equivalent to the point topos, whose point is the empty set. 2:44:48

Example. The open B^* classifies the objects x such that $x \to e$ is an epimorphism; its points are the nonempty sets. 2:46:57

Example. The empty subtopos classifies the theory that has a model only in the empty topos. 2:49:05

Example. The subtopos Φ_{01} classifies the subobjects of the terminal object ($x \to e$ a monomorphism); its points are the sets with at most one element. 2:50:10

Example. $\Phi_1 = \Phi_{01} \cap B^*$ classifies terminal objects; Φ_0 and Φ_1 are equivalent to the point topos but are two distinct points of B. 2:52:22

Example. $\Phi_{01} \simeq \widehat{\Delta_1}$, where $\Delta_1 = \{0 \to 1\}$: the morphisms $\mathcal{E} \to \widehat{\Delta_1}$ correspond to the monomorphisms $x \hookrightarrow e$. 2:55:37

Example. $\widehat{\Delta_1}$ is the topos of the Sierpiński space (spectrum of a discrete valuation ring); a morphism to it is equivalent to an open of the source topos. 2:57:45

Proposition. *Every T_0 topological space embeds in a product of copies of the Sierpiński space.* 2:58:47

Proposition. *The subtopoi of a topos form a complete lattice (arbitrary intersections and suprema).* 2:59:53

92 Flat functors and torsors

Example. In the opposite of finitely presented groups, the universal group is $\mathbb{Z}$ with its cogroup structure; the classifying topos of groups is its topos of presheaves. 3:06:38

Proposition. *$\widehat{C}$ classifies the functors $C \to \mathcal{E}$ whose continuous extension to $\widehat{C}$ is left exact (flat functors); if C has finite limits, these are the left exact functors.* 3:10:03

Proposition. *For a group G viewed as a category C with one object, $\widehat{C} = B_G$; a functor $C \to \mathcal{E}$ is an object X with a G-action, extended by $P \mapsto P \wedge^G X$.* 3:12:13

Definition. Contracted product: $P \wedge^G X$ is the quotient of $P \times X$ by the diagonal action of G, $(pg, x) \sim (p, gx)$. 3:13:21

Proposition. *This extension is left exact iff X is a torsor under the constant group $G_{\mathcal{E}}$: B_G classifies G-torsors.* 3:14:27

93 The classifying topos of a group

Tape 3, side B

Question. For a group G of a topos S, represent on topoi X over S the 2-functor of torsors under the inverse image of G. 3:15:58

Proposition. *The objects of S equipped with a left action of G form a topos B_G: the forgetful functor $B_G \to S$ is faithful and commutes with all limits.* 3:22:20

Proposition. *$p^* : S \to B_G$ equips an object with the trivial action; a group morphism $G \to G'$ induces by restriction an essential morphism $B_G \to B_{G'}$.* 3:24:28

Definition. Universal torsor: G acting on itself by left translations (an object of B_G), with the right action of $\overline{G} = p^*G$ by right translations. 3:26:37

Proposition. *Universality: for $q : X \to S$, the q^*G-torsors on X correspond to the morphisms $X \to B_G$ over S, via the contracted product.* 3:33:18

94 Classifying topoi: further examples

Discussion. A question on what structure the classifying topos of a monoid, rather than a group, would classify: the lecturer does not know, and instead explains torsors and locally free modules as the case of interest. 3:36:34

Proposition. *On a stack, the category of objects locally isomorphic to a section s, with their isomorphisms, is equivalent to that of torsors under $\underline{\mathrm{Aut}}(s)$.* 3:39:47

Example. Locally free $\mathcal{O}_X$-modules of rank n, with their isomorphisms, are equivalent to torsors under $\mathrm{GL}_n(\mathcal{O}_X)$. 3:40:51

Proposition. *The topos B_{GL_n} over the classifying topos of rings classifies ringed topoi equipped with a locally free module of rank n (with isomorphisms).* 3:42:59

Proposition. *Local rings are classified by the subtopos defined by the Zariski topology on affine schemes of finite type over $\mathbb{Z}$.* 3:44:02

Proposition. *Strictly local rings are classified by a smaller subtopos, defined by a finer topology (the étale topology).* 3:45:06

Counterexample. Locally constant objects do not come from a representable theory: they are not stable under colimits (totally disconnected spaces). 3:47:12

Counterexample. For $n \geq 2$, covering spaces of rank n of the terminal object do not form an algebraic structure: in **Set**, sets with n elements are not stable under filtered colimits. 3:48:16

Proposition. *With only isomorphisms as morphisms, the objects locally isomorphic to the constant object I_n with n elements are equivalent to torsors under the constant group $\mathfrak{S}_n$.* 3:50:26

Proposition. *The topos $B_{\mathfrak{S}_n}$ of $\mathfrak{S}_n$-sets classifies covering spaces of rank n of the terminal object, with their isomorphisms.* 3:51:31

95 Connected rings

Proposition. *Connectedness of the spectrum of a ring can be written in any topos (every idempotent is 0 or 1, and $0 \neq 1$) and defines a subtopos of the classifying topos of rings.* 3:52:37

Counterexample. For $n \geq 2$, "having exactly n connected components" does not define a theory: the property is not stable under filtered colimits. 3:53:42

Proposition. *A ring locally with n connected components is equivalent to a pair: I locally constant of rank n, and a 0-connected ring A_0 over I, of which A is the direct image.* 3:57:59

Proposition. *The functor $(I, A_0) \mapsto A$ is fully faithful if the ring morphisms taken are those inducing isomorphisms on the π_0.* 4:00:09

Remark. The classifying topos of rings locally with n components is constructed over $B_{\mathfrak{S}_n}$, by classifying a 0-connected ring over the universal covering space I_n. 4:03:21

Tape 4, side A

Proposition. *The classifying topos of a ring over an object I of a topos B exists; if I is a sum of n copies of the terminal object, it is $R^n \times B$, R the classifying topos of rings.* 4:09:38

96 Left exact functors and topoi

Proposition. *If $\mathcal{E}$ is a topos and C has finite limits, $C \to \mathcal{E}$ is left exact iff its extension $\widehat{C} \to \mathcal{E}$ commuting with colimits is.* 4:13:58

Proposition. *Converse: if $\mathcal{E}$ has a small strictly generating subcategory C stable under finite limits and this property holds, $\mathcal{E}$ is a topos.* 4:15:02

Proposition. *The classifying topos of a theory defined by finite limits is connected and acyclic.* 4:18:14

Definition. Elementary topos: a category with finite limits, cartesian closed, equipped with a subobject classifier. 4:22:31

Counterexample. Faithfulness of a G-action on X is not stable under inverse image: it is not an algebraic structure in the sense of topoi. 4:25:44

97 Topoi without points

Discussion. Can topological spaces be seen as structures inside a topos? A suggestion based on the power-set object is discussed; the lecturer is not convinced that it behaves well with inverse limits. 4:26:49

Proposition. *A theory represented by a nonempty topos may have no set-theoretic model: there exist topoi without points.* 4:35:21

Discussion. Structures classified by toposes need not have set-theoretic realizations: the lecturer explains that toposes without points exist, and a listener recalls an example built from a measure space. 4:35:21

Example (Deligne). Borel sets of $[0,1]$ modulo null measure, countable coverings up to null measure; the resulting topos has no point. 4:36:27

98 The classifying topos of fields

Proposition. *Classifying topos of fields: affine schemes of finite type over $\mathbb{Z}$, covering families generated by $\{D(f) \to X,\ V(f) \to X\}$ and by $X_{\mathrm{red}} \to X$.* 4:40:53

Proposition. *For X quasi-compact and quasi-separated, the constructible topology (opens: unions of constructible subsets) makes X compact, with the same points.* 4:41:59

Proposition. *For $X = \operatorname{Spec} A$, the colimit of the rings of functions of the finite constructible partitions (reduced structures) has as spectrum X with the constructible topology, with stalks the residue fields.* 4:43:01

Proposition. *The classifying topos of fields is identified with presheaves on these compact schemes (constructible morphisms) turning finite sums into products; the universal field is the ring of functions.* 4:44:06

99 Flatness

Definition. Absolutely flat rings: rings all of whose modules are flat; these are the reduced rings of Krull dimension zero. 4:45:11

Proposition. *Flatness of a module over a ring of a topos is stable under inverse image by a morphism of topoi (Deligne).* 4:47:21

Proposition. *Criterion: M is flat over A if for every U and every finitely generated ideal $J \subset A_U$, $M_U \otimes J \to M_U$ is a monomorphism.* 4:48:27

Proposition. *This criterion is a countable family of axioms: for each n, on $U_n = A^n$, the ideal generated by the n universal projections.* 4:49:31

Tape 4, side B

Discussion. When a notion such as flatness or connectedness has several candidate extensions to toposes, the lecturer says he discards any not stable under inverse image, and so far has met no serious ambiguity. 4:53:55

Remark. For a relative scheme of finite presentation over a ringed topos, two notions of flatness (Hakim): a flat morphism of ringed topoi, or locally arising from a flat morphism of schemes. 4:55:00

100 Hypertopoi

Proposition. *In* **Set**, *an arbitrary product of epimorphisms is an epimorphism (analogue of total disconnectedness); this does not hold in every topos.* 5:00:19

Proposition. *The morphism $\mathcal{E} \to \mathrm{pt}$ is essential iff the constant sheaf functor commutes with arbitrary products (analogue of local connectedness).* 5:01:24

Definition. The topoi $\widehat{C}$ have these two properties; that they are the only ones is stated without proof. "Hypertopos": these topoi, with essential morphisms; the notion is not defined more precisely. 5:02:29

Proposition. *A small Karoubian category D is determined by $\widehat{D}$: it is the category of its essential points.* 5:06:51

Proposition. *The category of points of $\widehat{C}$, C with finite limits, is* $\mathrm{Hom}_{\mathrm{lex}}(C, \mathbf{Set}) \simeq \mathrm{Ind}(C^{\mathrm{op}})$. 5:10:04

Remark. For hypertopoi, the only candidate classifying object of T_C is the topos of presheaves on the category of points of $\widehat{C}$, not small in general ("pseudo-hypertopos"). 5:11:18

101 Universes

Remark. A letter reports that Deligne has proved the Weil conjectures. The proof is not yet known to the lecturer, who expects it to use étale cohomology and hence universes. 5:15:36

Proposition. *The universe axiom is equivalent to: every cardinal is bounded by a strongly inaccessible cardinal; the existence of such a cardinal is not provable in ZFC, if ZFC is consistent.* 5:16:37

Remark. Asked whether the universe axiom could be shown consistent relative to ZFC, the lecturer recalls having been told by logicians, ten years earlier, that there was no hope. This does follow from Gödel's second incompleteness theorem, since the axiom implies the consistency of ZFC. 5:16:37

Remark. Samuel's idea of "galaxies", sets smaller than universes in which the union axiom is weakened, is recalled as a way to avoid a new axiom. The description given is unclear, and it is not reproduced here as a definition. 5:17:41

102 The standard conjectures

Discussion. In a discussion of foundations and the continuum hypothesis, a listener recalls an abelian group statement settled by that hypothesis, and the conversation turns to whether such questions have a definite answer. 5:25:07

Remark. The standard conjectures are formulated on algebraic cycles modulo numerical equivalence, with no particular cohomology; they include the equality of homological and numerical equivalence. 5:32:49

Remark. The standard conjectures would allow one to construct motives from algebraic cycles alone, independently of any cohomology. 5:33:55

Discussion. Asked for an example of a site that is not small, the lecturer proposes all schemes with the étale or flat topology, since some schemes are too big to be covered by a set. 5:53:34

Discussion. Subtopoi of a presheaf topos correspond to topologies on the category; as classifying topoi of structures, they correspond to adding axioms to a type of algebraic structure. 6:05:25

Discussion. Quotient topoi are suggested for foliations, where the naive quotient space may be very bad; the lecturer says a good quotient topos always exists, and points to exercises in SGA 4. 6:09:46

References

Numbers in square brackets refer to this list. It gathers what the lecturer tells the audience to read, with the moment he says it (video/timestamp), and the other sources used. A "?" marks a title the captions do not make certain.

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[3] M. Artin, A. Grothendieck, J.-L. Verdier, *SGA 4, Théorie des topos et cohomologie étale des schémas*, LNM 269, 270, 305, Springer, 1972–1973: exp. I on universes, generators and ind-objects, § 9 on accessible functors, which he says is told "somewhat inadequately" there (1/4, 2:59:16, 9:59:51; 3/4, 3:45:13, 4:24:37, 4:53:40); exp. V for cohomology (1/4, 7:16:42); the exercises (1/4, 8:16:56, 9:08:54).

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